Fast and Robust Simulation-Based Inference With Optimization Monte Carlo

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Abstract

Bayesian parameter inference for complex stochastic simulators is challenging due to intractable likelihood functions. Existing simulation-based inference methods often require large number of simulations and become costly to use in high-dimensional parameter spaces or in problems with partially uninformative outputs. We propose a new method for differentiable simulators that delivers accurate posterior inference with substantially reduced runtimes. Building on the Optimization Monte Carlo framework, our approach reformulates stochastic simulation as deterministic optimization problems. Gradient-based methods are then applied to efficiently navigate toward high-density posterior regions and avoid wasteful simulations in low-probability areas. A JAX-based implementation further enhances the performance through vectorization of key method components. Extensive experiments, including high-dimensional parameter spaces, uninformative outputs, multiple observations and multimodal posteriors show that our method consistenly matches, and often exceeds, the accuracy of state-of-the-art approaches, while reducing the runtime by a substantial margin.

1 Introduction

Parametric stochastic simulators are widely used to model complex processes across biology, physics, health sciences, and many other disciplines. As simulators become more accurate, they also become more predictive. However, accuracy comes with greater complexity, such as a large numbers of free parameters or a high-dimensional output. This complexity, in turn, poses significant challenges for parameter inference.

A variety of methods, collectively referred to as Likelihood-Free (LFI) or Simulation-Based (SBI) Inference (Lintusaari et al., 2017; Sisson et al., 2018; Cranmer et al., 2020), have been developed to infer simulator parameters. These include sampling-based techniques, like Approximate Bayesian Computation (ABC, Marin et al., 2012), and approaches based on summary statistics (Chen et al., 2021, 2023), parametric modeling (Wood, 2010; Price et al., 2018), and ratio estimation (Thomas et al., 2022; Durkan et al., 2020). More recently, substancial progress has been driven by neural methods, which use deep networks to learn informative summary statistics (Radev et al., 2020) and to estimate either the posterior (Greenberg et al., 2019; Papamakarios and Murray, 2016) or the likelihood (Papamakarios et al., 2019) function.

Neural-based approaches have greatly broadened the applicability of SBI to complex problems. However, they remain highly data-intensive, requiring large volumes of simulated datasets to train their neural estimators. Since both data generation, using simulation calls, and neural network training are computationally costly, these methods often incur significant runtimes, which in turn restricts experimental flexibility and hinders practical deployment.

These challenges are particularly pronounced in two scenarios. First, in high-dimensional parameter spaces, SBI methods face the curse of dimensionality, as they rely on densely populating both parameter and data spaces with samples (Lintusaari et al., 2017; Sisson et al., 2018; Cranmer et al., 2020). Second, in distractor settings, where complex simulators exhibit intricate interactions between parameters and outputs, some outputs may depend only weakly, or not at all, on certain parameters (Saltelli et al., 2008; Monsalve-Bravo et al., 2022). Such uninformative outputs act as "distractors", complicating inference (Lueckmann et al., 2021). As shown in Figure 1, accurate inference with existing neural-based approaches comes only at the

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cost of many simulations and substantial runtimes.

We propose a new LFI method that uses gradient information to overcome these challenges. Building on the Robust Optimization Monte Carlo framework (ROMC, Ikonomov and Gutmann, 2020), we reformulate the stochastic simulation as a set of deterministic optimization problems and use gradient descent to obtain posterior samples. Gradients, as in standard optimization, provide efficient navigation in high-dimensional parameter spaces, so they rapidly guide us toward high-density posterior regions. Moreover, they enable sensitivity analyses which allows us to identify and filter out distractor dimensions, thereby improving inference quality.

The ROMC framework provides the theory for casting inference as deterministic optimization, but, it has two key limitations; it lacks a strategy for defining an appropriate distance function in problems with distractors or multiple iid observations, and its latest implementation (Gkolemis et al., 2023) does not exploit gradients or other computational accelerations, which limits its applicability to low-dimensional problems. We overcome these challenges and present R2OMC, an SBI method that when the simulator is differentiable,

- enables posterior sampling in high-dimensional parameter spaces under a limited simulation budget,
- automatically identifies and filters out uninformative dimensions (distractors),
- handles-well multiple iid observations,
- runs efficiently through a `JAX` implementation that uses automatic differentiation, vectorization, and just-in-time compilation.

R2OMC is systematically compared to state-of-the-art SBI methods, demonstrating accurate posterior inference at a fraction of the computational cost in a wide range of scenarios.

2 Background and Motivation

Neural-based SBI methods are data-intensive, with their accuracy depending on access to large simulated datasets. This leads to two main sources of computational cost: (a) generating datasets from complex simulators, and (b) training deep neural estimators on high-volume data. Prior work (Lueckmann et al., 2021) has noted that training is often the more time-consuming step. Further details are provided in Appendix B. Moreover, with modern frameworks, such as JAX, which enable extensive vectorization, simulation can be accelerated, further shifting the bottleneck toward training. This motivates us to search for a method that sidesteps the bottleneck of separate neural estimator training.

2.1 ROMC framework

ROMC has been introduced by Ikonomov and Gutmann (2020), extending and addressing robustness issues of the Optimization Monte Carlo method by Meeds and Welling (2015). It belongs to the larger family of ABC methods that approximate the likelihood function as the probability of simulating an output ϵ-close to the observation,

$$L_\epsilon(\boldsymbol{\theta}) = \Pr(d(\mathbf{y}, \mathbf{y}^o) \leq \epsilon \mid \boldsymbol{\theta}). \tag{1}$$

Here, $d(\cdot, \cdot)$ is a distance, $\mathbf{y}$ the simulator output, and $\mathbf{y}^o$ the observed data. In the following, we denote by $g(\boldsymbol{\theta}, \mathbf{u})$ the simulator that maps noise (nuisance) variables $\mathbf{u} \sim p(\mathbf{u})$[1] and parameters $\boldsymbol{\theta}$ to multivariate data $\mathbf{y} = g(\boldsymbol{\theta}, \mathbf{u})$. Introducing the acceptance region $C_\epsilon = \{(\boldsymbol{\theta}, \mathbf{u}) : d(g(\boldsymbol{\theta}, \mathbf{u}), \mathbf{y}^o) \leq \epsilon\}$ and using the law of the unconscious statistician, (1) can be written as

$$L_\epsilon(\boldsymbol{\theta}) = \mathbb{E}_{p(\mathbf{u})}\left[\mathbb{1}_{C_\epsilon}(\boldsymbol{\theta}, \mathbf{u})\right], \tag{2}$$

where $\mathbb{1}_{C_\epsilon}$ denotes the indicator function that is one if $(\boldsymbol{\theta}, \mathbf{u}) \in C_\epsilon$ and zero otherwise. When approximating the expectation as a sample average, we obtain

$$L_\epsilon(\boldsymbol{\theta}) \approx \frac{1}{S} \sum_{i=1}^{S} \mathbb{1}_{C_\epsilon^i}(\boldsymbol{\theta}), \tag{3}$$

which is the proportion of acceptance regions $C_\epsilon^i = \{\boldsymbol{\theta} : d(g(\boldsymbol{\theta}, \mathbf{u}_i), \mathbf{y}^o) \leq \epsilon\}$ that contain $\boldsymbol{\theta}$.

ROMC takes (3) as starting point and focuses on characterizing the acceptance regions C_ϵ^i, because once the C_ϵ^i are known, we can not only evaluate the approximate likelihood function, but also obtain posterior samples efficiently. ROMC introduces uniform proposal distributions q_i with support on C_ϵ^i only. Samples from $\boldsymbol{\theta}_{ij} \sim q_i$ for $j \in 1, \ldots, M$, are then equal to posterior samples if weighted with the (unnormalized) weights w_{ij}, $i = 1, \ldots, S, j = 1, \ldots, M$,

$$w_{ij} = \frac{p(\boldsymbol{\theta}_{ij})}{q_i(\boldsymbol{\theta}_{ij})} \mathbb{1}_{C_\epsilon^i}(\boldsymbol{\theta}_{ij}). \tag{4}$$

For the characterization of the acceptance regions C_ϵ^i, ROMC first determines the minimizers $\boldsymbol{\theta}_i^*$,

$$\boldsymbol{\theta}_i^* = \text{argmin}_{\boldsymbol{\theta}} d_i(\boldsymbol{\theta}), \quad d_i(\boldsymbol{\theta}) = d(g(\boldsymbol{\theta}, \mathbf{u}_i), \mathbf{y}^o), \tag{5}$$

where $d_i(\boldsymbol{\theta})$ are deterministic distance functions because the sources of randomness (seed) are set to $\mathbf{u}_i$

[1] In a computer program, sampling $\mathbf{u}$ corresponds to sampling a random seed.

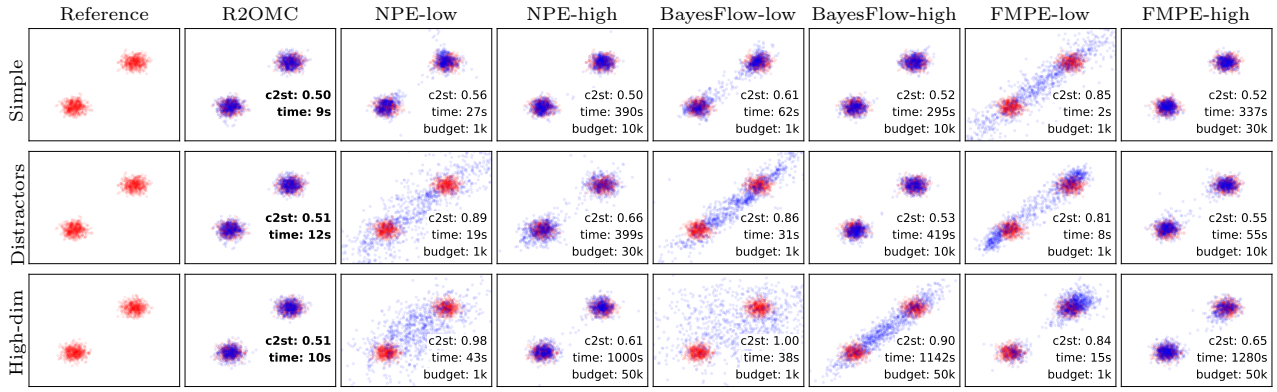


Figure 1: Top to bottom: Simple, high-dimensional, distractors, multiple observations, combined. Left to right: ground truth, NPE, NLE, Naive-ROMC, R2OMC (proposed). Details in Section C of the Appendix.

and kept fixed. Next, it starts from the optimization end points $\boldsymbol{\theta}_i^*$ to map out C_ϵ^i using for ϵ a value derived from the quantiles of the minimial distance functions, $d_i^* = d_i(\boldsymbol{\theta}_i^*), i = 1, \dots, S$ (see Ikonomov and Gutmann, 2020, for details). While several options are possible (Ikonomov and Gutmann, 2020), we here use hyper-boxes centered at $\boldsymbol{\theta}_i^*$, which can be constructed with a line search algorithm for each parameter dimension. For details, please see Appendix A.

The power of the ROMC framework stems from the possibility to use optimization, in particular gradient-based methods, to determine the optimization end points $\boldsymbol{\theta}_i^*$, and hence the acceptance regions C_ϵ^i and the proposal distributions q_i. We will exploit this in our method to scale SBI to higher parameter dimensions. Before we can tap into the ROMC framework, however, we will need to overcome challenges related to the definition of the distance function d in case of multiple iid observations and the presence of distractors.

2.2 ROMC challenges and limitations

ROMC suffers from three main limitations; (a) it fails in the presence of distractors, (b) it does not provide a strategy for handling multiple iid observations, and (c) its existing implementation does not use gradients and other computational accelerations, so it can only be applied to low-dimensional problems.

Uninformative dimensions (distractors). Distractors are output dimensions that are unrelated to the parameters of interest. For example, consider a graphics renderer where the parameter of interest controls a scene element, e.g. the appearance of vase, so that most pixel values (output dimensions) are unrelated to the parameter and thus serve as distractors. To emulate distractors consider the simulator of Figure 1 (second row, more details in Appendix C). There, the simulator can be written as $\mathbf{y} = (\mathbf{y}^{(1)}, \mathbf{y}^{(2)})$, where the first set of dimensions is $\mathbf{y}^{(1)} \sim 0.5\,(\mathcal{N}(\boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \mathbf{I}))$, and the second $\mathbf{y}^{(2)} \sim \mathcal{N}(\boldsymbol{\mu}, 100\mathbf{I})$, where $\boldsymbol{\mu} \sim \mathcal{U}(-\mathbf{10}, \mathbf{10})$. In this setup, only $\mathbf{y}^{(1)}$ depends on $\boldsymbol{\theta}$, while $\mathbf{y}^{(2)}$ emulates distractors, so, when we observe $\mathbf{y}^o = (\mathbf{y}^{o,(1)}, \mathbf{y}^{o,(2)})$, the true posterior depends solely on $\mathbf{y}^{o,(1)}$. ROMC is vulnerable to failure in this setting because typical distance functions d consider all output dimensions equally. Therefore, the minimal distances d_i^* will generally be large for seeds that generate $\mathbf{y}^{(2)}$ significantly different from $\mathbf{y}^{o,(2)}$. This results in an inflated ϵ, and in turn to a much broader posterior than the ground truth.

Multiple iid observations. In many applications, we have access to multiple iid observations $\{\mathbf{y}_n^o\}_{n=1}^N$. For instance, in neuroscience, researchers often collect multiple recordings of the same neural response to a stimulus. ROMC does not prescribe a clear strategy for dealing with such cases. A straightforward approach would be to run the simulator with a different seed $\mathbf{u}_{i,n}$ for each observation n and then average the distances: $d_i(\boldsymbol{\theta}) = 1/N \sum_n \tilde{d}(g(\boldsymbol{\theta}, \mathbf{u}_{i,n}), \mathbf{y}_n^o)$, where $\tilde{d}$ is the distance for a single data point. However, this approach is sensitive to the ordering of the observations; the distance increases or decreases artificially if the observations are permuted. As before, this will generally result in artificially large minimal distances d_i^*, in an excessively large ϵ and to a much broader posterior than the ground truth. Other approaches, such as concatenating all observations into a single vector, encounter similar issues. Note that for squared Euclidean distances, the averaging and concatenation approaches are mathematically equivalent.

Computational efficiency. Current ROMC implementations (Gkolemis et al., 2023) suffer from significant computational bottlenecks that restrict their applicability to low-dimensional problems. First, opti-

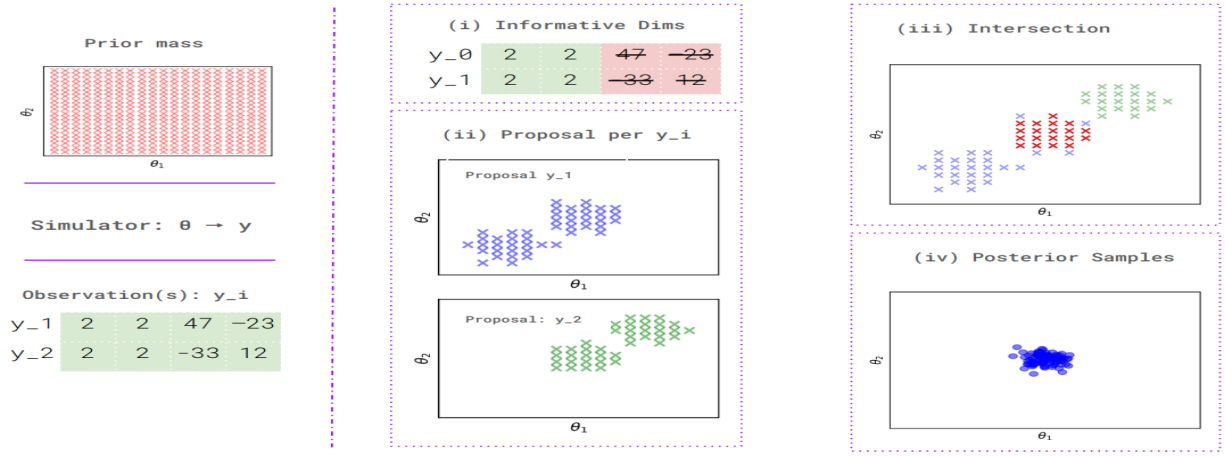


Figure 2: Overview R2OMC: Distractors are automatically filtered out and proposal distributions q_i^n are created for each observation (here two): the blue-dotted region indicates where the proposal distributions for the first observation are nonzero, and the green-dotted region indicates the same for the second observation. Only samples that fall within the intersection of both regions are given a positive weight and become posterior samples.

mization relies on gradient-free methods, e.g., Bayesian optimization, which scale poorly with the number of parameters and require many simulator evaluations. Second, the construction of the acceptance regions C_ϵ^i with a line-search algorithm (see Appendix A), require repeated simulator evaluations per seed and per dimension, which also scales poorly with the parameter dimension.

3 Proposed method: R2OMC

We here present our new method for Bayesian parameter inference in simulator models. The method uses gradients to obtain posterior samples even in high-dimensional parameter spaces. The method belongs to the ROMC framework so that we start with discussing how we deal with iid observations and distractors before discussing its implementation.

3.1 Uninformative dimensions (distractors)

Distractors can negatively affect the quality of the posterior approximation. To counteract their influence, we measure the sensitivity of the output dimensions with respect to the parameters $\boldsymbol{\theta}$ by computing the average norm of their derivative with respect to $\boldsymbol{\theta}$, and checking that the value is above a minimal value (threshold) τ,

$$I_i = \mathbb{E}_{p(\boldsymbol{\theta})p(\mathbf{u})} \left[||\nabla_{\boldsymbol{\theta}} g_i(\boldsymbol{\theta}, \mathbf{u})|| \right] > \tau. \tag{6}$$

The expectation equals the expectation with respect to the joint distribution of the prior and the i-th dimension of the simulator output, $g_i(\boldsymbol{\theta}, \mathbf{u})$. In our experiments, we approximate the expectation using 50 samples from $p(\boldsymbol{\theta})$ and $p(\mathbf{u})$ each. Collecting all binary indicator variables I_i into the vector $\mathbf{m}_\tau$, we can filter out uninformative dimensions by computing the distance between observed and simulated data for informative dimensions only. To compute $\boldsymbol{\theta}_{i,n}^*$ in (11), we minimize

$$d_i^n(\boldsymbol{\theta}) = d(g(\boldsymbol{\theta}, \mathbf{u}_{i,n}) \odot \mathbf{m}_\tau, \mathbf{y}_n^o \odot \mathbf{m}_\tau). \tag{7}$$

The procedure is illustrated in Figure 2, step (i).

The threshold τ controls the sensitivity of the test. In our experiments, we set τ to machine precision; more sophisticated approaches consist in monitoring the distribution of the expected norm of the gradient, or by assessing false vs true positives under control conditions. This is possible since the mask $\mathbf{m}_\tau$ can be computed prior to receiving observed data.

3.2 Multiple iid observations

The likelihood for multiple iid observations $\{\mathbf{y}_n^o\}_{n=1}^N$ is the product of the likelihoods for each data point. Approximating the likelihood for each data point as in (3) and multiplying-in the prior $p(\boldsymbol{\theta})$, we can obtain a formal expression for the approximate posterior

$$p_\epsilon(\boldsymbol{\theta}|\{\mathbf{y}_n^o\}) \propto p(\boldsymbol{\theta}) \prod_n^N \sum_i^S \mathbb{1}_{C_\epsilon^{i,n}}(\boldsymbol{\theta}), \tag{8}$$

where $C_\epsilon^{i,n}$ is the acceptance region for data point n and source of randomness (seed) $\mathbf{u}_{i,n}$,

$$C_\epsilon^{i,n} = \{\boldsymbol{\theta} : d(g(\boldsymbol{\theta}, \mathbf{u}_{i,n}), \mathbf{y}_n^o) \leq \epsilon\}. \tag{9}$$

We have S sources of randomness for each data point $\mathbf{y}_n^o$, and so in total $S * N$ acceptance regions $C_\epsilon^{i,n}$.

We now address the question how to efficiently sample from $p_\epsilon(\boldsymbol{\theta}|\{\mathbf{y}_n^o\})$. To achieve this, we consider the task of computing the posterior expectation $\bar{h} = \mathbb{E}_{p_\epsilon}[h(\boldsymbol{\theta})]$ under $p_\epsilon(\boldsymbol{\theta}|\{\mathbf{y}_n^o\})$. Plugging in the expression in (8) gives, after normalization,

$$\bar{h} = \frac{\int h(\boldsymbol{\theta}) p(\boldsymbol{\theta}) \prod_n^N \sum_i^S \mathbb{1}_{C_\epsilon^{i,n}}(\boldsymbol{\theta}) d\boldsymbol{\theta}}{\int p(\boldsymbol{\theta}) \prod_n^N \sum_i^S \mathbb{1}_{C_\epsilon^{i,n}}(\boldsymbol{\theta}) d\boldsymbol{\theta}}. \tag{10}$$

The integrals in the equation are typically intractable. But they correspond to expectations with respect to the prior $p(\boldsymbol{\theta})$ and hence could be approximated with a sample average. However, this is very inefficient. Few (or no) prior samples will generate outputs ϵ-close to all observations, so most (or all) will get zero weight, leading to a very low effective sample size. Following the general ROMC framework by Ikonomov and Gutmann (2020), we thus employ importance sampling with an adaptive proposal distribution obtained via optimization.

We first treat each data point $\mathbf{y}_n^o$ separately and construct proposal distributions $q_i^n(\boldsymbol{\theta})$, $i = 1, \ldots, S$, as in Section 2.1: We use optimization to find the minimizers $\boldsymbol{\theta}_{i,n}^*$ of the deterministic functions $d_i^n(\boldsymbol{\theta}) = d(g(\boldsymbol{\theta}, \mathbf{u}_{i,n}), \mathbf{y}_n^o)$,

$$\boldsymbol{\theta}_{i,n}^* = \text{argmin}_{\boldsymbol{\theta}} d_i^n(\boldsymbol{\theta}), \tag{11}$$

and then build a hyperbox around each acceptance region $C_\epsilon^{i,n}$ and define $q_i^n(\boldsymbol{\theta})$ to be the uniform distribution over it. The optimization problem in (11) is deterministic and we solve it with gradient-based methods to enable scalability to large parameter spaces. Note that treating each data point $\mathbf{y}_n^o$ separately allows us to avoid the issues explained in Section 2.2.

We propose to combine the per-data point proposal distributions $q_i^n(\boldsymbol{\theta})$ in form of a mixture distribution,

$$q(\boldsymbol{\theta}) = \frac{1}{N}\frac{1}{S}\sum_{n=1}^{N}\sum_{i=1}^{S} q_i^n(\boldsymbol{\theta}). \tag{12}$$

The posterior expectation in (10) can then be approximated in form of a weighted average

$$\bar{h} = \frac{\sum_p w_p h(\boldsymbol{\theta}_p)}{\sum_p w_p}, \quad \boldsymbol{\theta}_p \sim q(\boldsymbol{\theta}) \tag{13}$$

where the weights w_p, $p = 1, \ldots, P$, are given by

$$w_p = \frac{p(\boldsymbol{\theta}_p)}{q(\boldsymbol{\theta}_p)} \prod_{n=1}^{N}\sum_{i=1}^{S} \mathbb{1}_{C_\epsilon^{i,n}}(\boldsymbol{\theta}_p). \tag{14}$$

This means that the $\boldsymbol{\theta}_p \sim q(\boldsymbol{\theta})$, weighted by w_p, represent samples from the posterior $p_\epsilon(\boldsymbol{\theta}|\{\mathbf{y}_n^o\})$ in (8).

The rationale for choosing $q(\boldsymbol{\theta})$ in (12) as proposal distribution is as follows: it is a mixture of uniform distributions, so sampling and evaluating is easy. It seamlessly combines the contribution from multiple data points, allowing future data points to be included without re-computation. If each q_i^n encloses the corresponding acceptance region $C_\epsilon^{i,n}$, then $q(\boldsymbol{\theta}) > 0$ in areas of the parameter space where $\prod_n^N \sum_i^S \mathbb{1}_{C_\epsilon^{i,n}}(\boldsymbol{\theta}) > 0$ and no posterior mass is discarded in the weighting in (14). Finally, regions of high likelihood are characterized by regions where multiple $C_\epsilon^{i,n}$ overlap. As the proposal distribution $q(\boldsymbol{\theta})$ consists of a mixture of the $q_i^n(\boldsymbol{\theta})$ with equal weights, areas where multiple $C_\epsilon^{i,n}$ overlap will be sampled more often, so that areas of high likelihood will be represented with more samples.

Figure 2, steps (ii-iii), illustrates the proposed procedure to obtain posterior samples. We have two observed data points and the $\boldsymbol{\theta}_p$ come from regions that match at least one observation; the blue-dotted area for $\mathbf{y}_1^o$ and the green-dotted region for $\mathbf{y}_2^o$. However, only the ones that match both will get a positive weight (red dots) and become posterior samples.

3.3 `JAX` implementation

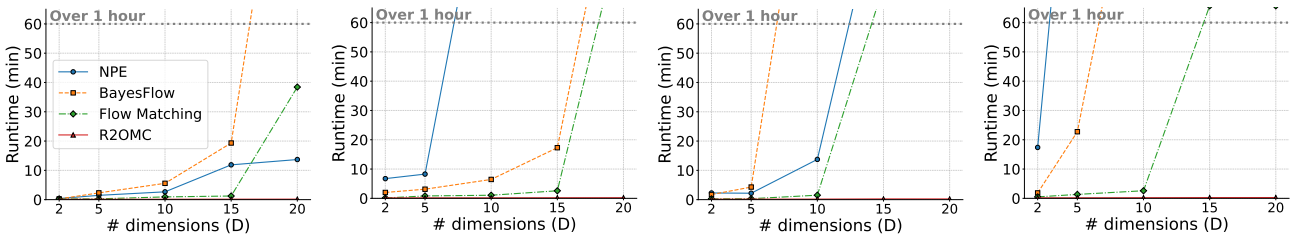
Algorithms 1 summarizes the key steps of R2OMC. It has three steps: dealing with distractors (line 2); constructing per-data point proposal distributions $\{q_i^n\}_{i=1}^S$ (lines 4-5); combining them to form the overall proposal distribution $q(\boldsymbol{\theta})$ and generating the weighted samples (lines 7-8).

Algorithm 1 `R2OMC`. Requires: $p(\boldsymbol{\theta}), g(\boldsymbol{\theta}, \mathbf{u}), \{\mathbf{y}_n^o\}_{n=1}^N$

1: **procedure** R2OMC(τ, S, P)
2: $\quad\mathbf{m}_\tau = (I_1, \ldots, I_{D_y})$ using (6) with τ
3: $\quad$**for** $n \leftarrow 1$ to N **do**
4: $\quad\quad\boldsymbol{\theta}_{i,n}^* = \text{argmin}_{\boldsymbol{\theta}} d_i^n(\boldsymbol{\theta})$ using (7) for all i
5: $\quad\quad$Compute hyperboxes and $\{q_i^n\}_{i=1}^S$
6: $\quad$**end for**
7: $\quad\boldsymbol{\theta}_p \sim q(\boldsymbol{\theta})$, where $q(\boldsymbol{\theta}) = \frac{1}{N}\frac{1}{S}\sum_n \sum_i q_i^n(\boldsymbol{\theta})$
8: $\quad$Compute $\{w_p\}_p^P$ using (14)
9: $\quad$**return** $\{w_p, \boldsymbol{\theta}_p\}_{p=1}^P$
10: **end procedure**

We offer an efficient `JAX` implementation, benefiting from automatic differentiation (`grad`), vectorization (`vmap`), and just-in-time compilation (`jit`). In many cases, R2OMC executes in seconds where most SBI methods require minutes (see Section 4).

R2OMC requires approximately $N*S$ evaluations of the simulator: For the distractors, `grad` computes $\partial g_i/\partial\theta_j$ for all i and j in one call, and `vmap` evaluates multiple $\boldsymbol{\theta}$ at once. Combining them, the cost is S. Obtaining the per-data point proposal distributions $\{q_i^n\}_{i=1}^S$ consists



(a) $S_{\texttt{base}}$ (b) $S^{\texttt{dist}}_{\texttt{base}}$ (c) $S_{\texttt{MoG}}$ (d) $S^{\texttt{dist}}_{\texttt{MoG}}$

Figure 3: MoG benchmark: Success Frontier plots showing the lowest budget required to reach a C2ST score ≤ 0.75 for varying D.

of optimization (line 4) and hyperbox building (line 5) for each data point. Each of S optimizations requires T gradient descent steps, where T is a user-defined hyperparameter. `jit` allows compiling T_1 consequtive steps to apply them at the cost of a single execution, so the cost reduces to $S\frac{T}{T_1}$. $\frac{T}{T_1}$ is normally small, and can even reach 1 if memory allows setting $T_1 = T$. Hyperbox building also requires of the order of S steps when the required line searches are parallelized over each parameter dimensions using `vmap` (see Appendix A). Thus the total cost of this step is of the order of $N * S$. For generating the weighted samples, (14) is evaluated on P samples (line 8). The simulator is only used for computing $\prod_n^N \sum_i^S \mathbb{1}_{C_\epsilon^{i,n}}(\boldsymbol{\theta}_p)$ which requires evaluating $S * N$ different distance functions on P points each. Using `vmap` in each distance function, the total cost is $S * N$. Evaluating $q(\cdot)$ and $p(\cdot)$ does not require calling the simulator. Finally, if the simulator is expensive, the $S * N$ cost of checking whether the samples are in the acceptance regions can be avoided by using the hyperbox or another model for the acceptance region, see Ikonomov and Gutmann (2020).

The code is available at *anonymized link*.

3.4 Discussion - Limitations

While R2OMC provides efficient inference in many scenarios, it has several inherent limitations. First, it can only be applied to differentiable simulators, making it a gray-box approach rather than fully black-box. Second, its performance depends critically on the success of the underlying optimization: if the optimization fails, the resulting proposal samples may poorly approximate the posterior. Additionally, increasing the simulation budget does not necessarily translate into improved accuracy, highlighting that R2OMC's efficiency gains do not guarantee monotonic improvements with more resources.

Other limitations arise in specific inference scenarios. In the multiple observations setting, R2OMC optimizes with respect to each observation independently; if the resulting per-observation posteriors have little overlap, there is no guarantee that the combined samples accurately reflect the joint posterior. Finally, because the method's optimization does not explicitly account for the prior, proposal regions may occasionally fall in regions with low or no prior support, which can reduce inference quality. These limitations should be considered when applying R2OMC to complex or constrained SBI tasks.

4 Experiments

To assess R2OMC, we conduct experiments on a Mixture of Gaussians benchmark, on the synthetic SBI benchmark and an image-based inference problem, covering scenarios with high-dimesional data, distractors, multiple observations and complex posterior shapes.

Methods. In the MoG benchmark, we compare R2OMC against three well-known neural methods: NPE (Greenberg et al., 2019), BayesFlow (Radev et al., 2020) and FlowMatching (Wildberger et al., 2023). We use our `JAX` implementation for R2OMCand for the competitors the implementation provided by the `SBI Pytorch` package. In the SBI benchmark, we compare R2OMC against all methods reported by Lueckmann et al. (2021). In the image-based inference problem, we illustrate the applicability of R2OMC to a high-dimensional real-world problem, without comparing it. All experiments were conducted on a single Dell XPS PC equipped with an Intel® Core™ i7-10750H CPU @ 2.60GHz with 12 cores. Appendix D contains further details on the setup and methods.

Metrics. All methods are evaluated with the `C2ST` score (Gutmann et al., 2018; Lueckmann et al., 2021), which uses classification to measure the similarity between two sets; the first set are the samples from the approximate posterior and the second set from the ground truth. The best score is 0.5, indicating that the two sets are indistinguishable, while the worst score is 1.0, indicating perfect separation. As classifier, we

use a fully-connected neural network trained on $1,000$ samples from each set and report the mean C2ST score over 5 independent runs.

4.1 MoG benchmark

We systematically evaluate SBI methods when the dimensionality of the parameter space D increases. We use two simulators, $S_{\texttt{base}} : \mathbf{y} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2\mathbf{I})$ and $S_{\texttt{MoG}} : \mathbf{y} \sim \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta}, \sigma_1^2\mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \sigma_2^2\mathbf{I})\right)$, that lead to uni-modal and bi-modal posteriors respectively. For each simulator, we consider a simple version, where the output has the same dimensionality as the parameter vector ($D_y = D$), and a distractor version, where where we append 100 uninformative dimensions to the output ($D_y = D+100$). The distractor dimensions are sampled from $\mathcal{N}(\boldsymbol{\mu}, 100\mathbf{I})$, with $\boldsymbol{\mu} \sim \mathcal{U}(-10, 10)$. That is, we consider four simulators in total.

As observed data, we use $\mathbf{y}_n^o \approx (5, \ldots, 5)$, that is a vector of 5's with some small noise around it. In case of distractors, we append 100 zeros to it. The posterior for $S_{\texttt{base}}$ is then approximated by a truncated Gaussian, while the posterior for $S_{\texttt{MoG}}$ is well approximated by a truncated bi-modal Gaussian.

For each simulator, we evaluate R2OMC against the three neural methods mentioned above, varying D from 2 to 20 and the simulation budget from 1_000 to 100_000. For all simulators, we use $\mathcal{U}(-\mathbf{3}, \mathbf{3})$ as prior. For more details on the setup, see Appendix D.

Figure 3 summarizes the results in form of Success Frontier plots, showing the lowest budget required for a successful inference, defined as reaching a C2ST score ≤ 0.75, for varying D. All neural-based methods follow the same trend; as D increases, so does the requird budget, and soon they ask for more than 100_000 simulations to succeed, which incurs very high runtimes (see Appendix D). They struggle more with $S_{\texttt{MoG}}$ than with $S_{\texttt{base}}$, as the posterior is more complex and more on the distractor versions than on the simple ones. In contrast, R2OMC succeeds with a budget of up to 1_000 simulations and a runtime of a few seconds for all D and all simulators.

4.2 SBI benchmark

The SBI (Lueckmann et al., 2021) is a widely used benchmark for SBI methods. We select two characteristic experiments from the benchmark that cover different challenges.

4.2.1 SLCP (T.3) and plus distractors (T.4)

The SLCP (Simple Likelihood, Complex Posterior) task has a uniform prior over five parameters $\boldsymbol{\theta} \in \mathcal{U}(-\mathbf{3}, \mathbf{3})^5$ and four two-dimensional observations drawn from a

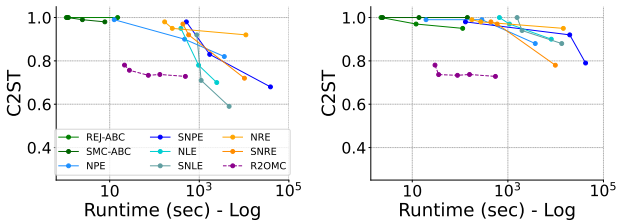


Figure 4: SLCP (left) and SLCP with distractors (right): C2ST score vs. runtime.

Gaussian distribution whose mean and variance are nonlinear functions of θ. This creates a posterior with four symmetric modes and sharp vertical cut-offs. SLCP with distractors augments the original SLCP task, adding 92 non-informative dimensions for a total of a 100D output.

Figure 4 shows C2STscore versus runtime for R2OMCand methods from Lueckmann et al. (2021) on both SLCP variants. R2OMC achieves accurate posterior approximation (C2ST score 0.7–0.8) in seconds using a budget of 10^3 samples. Other methods require significantly longer runtimes (up to hours) and budgets of at least 10^4 (often 10^5) samples for similar performance. Results are more pronounced with distractors, where most methods fail to reach C2ST scores below 0.8 regardless of budget.

Figure 5 illustrates R2OMCability to handle multiple observations by using the weighting scheme in R2OMC(Eq.14). The left panel shows pairwise posteriors using all proposal samples, i.e., the ones that are in ϵ-distance per observation, and represent a much broader distribution. Out of them, only the ones that are in ϵ-distance to all observations are given a positive weight and become posterior samples, as shown in the right panel.

4.2.2 Two-moons (T.8)

The 2-moons task tests whether the methods can estimate posteriors with bimodal and crescent shape structure. The simulator is $\mathbf{y}|\boldsymbol{\theta} \sim (r\cos(\alpha)+0.25, r\sin(\alpha)) + (-|\theta_1+\theta_2|/\sqrt{2}, (-\theta_1+\theta_2)/\sqrt{2})$ with $\alpha \sim \mathcal{U}(-\pi/2, \pi/2)$ and $r \sim \mathcal{N}(0.1, 0.01^2)$. The dimensionality is $D = 2$ and the prior is $\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{1}, \mathbf{1})$, creating a posterior with two half-moon shape modes.

Figure 6 shows the C2ST score and the pairwise posterior of R2OMC and all methods from Lueckmann et al. (2021). R2OMC achieves a C2STclose to 0.5 with a runtime of a few seconds (budget of 10^3 samples), whereas other methods require much longer runtimes (minutes to hours) to reach a similar score (budget of 10^5 samples). Additional results showing the scores vs. budget are in Appendix D. The pairwise poste-

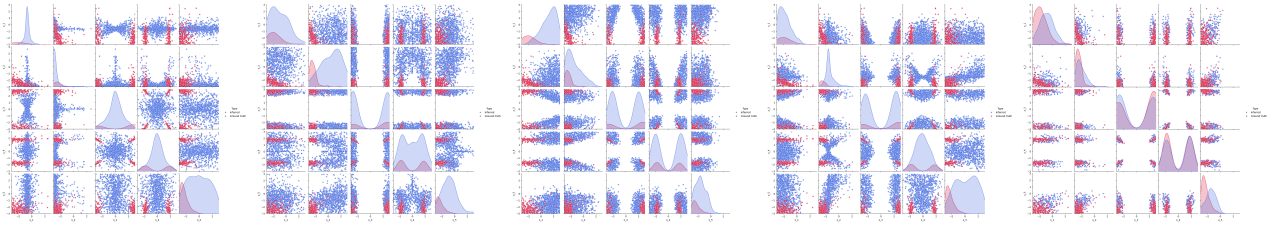


Figure 5: SLCP pairwise posteriors: using all proposal samples (left) and only the selected ones (right).

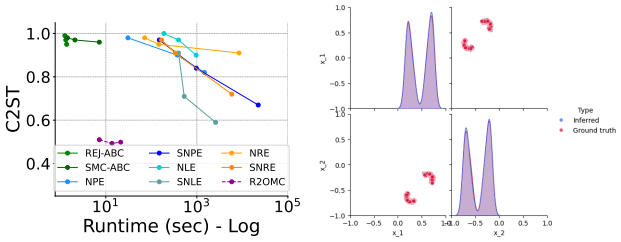


Figure 6: Two-moons results: C2ST score vs. runtime (Left) and pairwise posterior (Right).

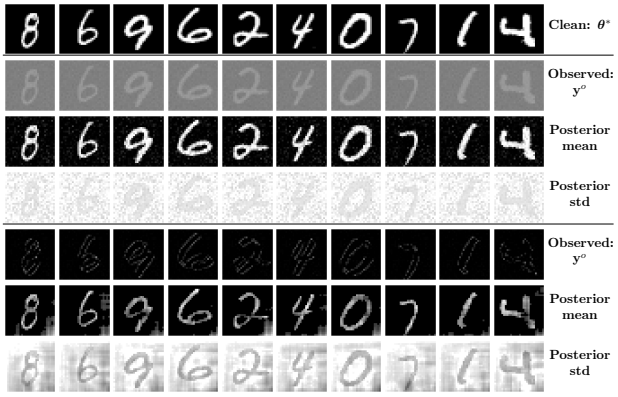


Figure 7: Image-based inference results. Top row shows the clean ground truth image $\boldsymbol{\theta}^*$. Rows 2–4 display the distorted observation, posterior mean, and posterior standard deviation obtained by R2OMCfor pixel-wise distortion. Rows 5–7 show the same results for the edge-detection filter distortion.

rior shows that R2OMC accurately captures the two crescent-shaped modes of the posterior.

4.3 Image-based inference

We evaluate R2OMC on (very) high-dimensional parameter spaces through image inference in noisy camera models. Two simulators are tested on MNIST dataset observations (LeCun et al., 1998). The first applies pixel-wise intensity changes, $y_{ij} = a\theta_{ij} + b + \boldsymbol{\epsilon}$, where θ_{ij} is pixel intensity and a and b control contrast and brightness. The second applies edge-detection filtering, $\mathbf{y} = \text{filter}(\boldsymbol{\theta}) + \boldsymbol{\epsilon}$, using convolution with filter$(\cdot)$. Both use Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, 0.1^2\mathbf{I})$ and uninformative prior $\boldsymbol{\theta} \sim \mathcal{U}(0,1)^{28\times 28}$.

Figure 7 shows that for both simulators, the posterior mean obtained by R2OMCvisually resembles to the original clean image, indicating that the posterior mode is close to the true generating parameter $\boldsymbol{\theta}^*$. SBI inference on images is highly difficult due to the curse of dimensionality. Few methods like GATSBI (Ramesh et al., 2022) succeed but require complex generative models with long, and often unstable, training. In contrast, R2OMC achieves accurate inference with only 100 simulations and a runtime of a few seconds.

5 Conclusion

We have presented R2OMC, a new Bayesian inference method for simulator models that addresses three key scalability challenges in simulation-based (likelihood-free) inference, namely high-dimensional parameter spaces, multiple observations, and the possible presence of uninformative dimensions (i.e., distractors). R2OMC builds on the ROMC framework and formulates the inference problem in terms of a set of deterministic optimization problems that can be solved with gradient-descent, enabling inference in high-dimensional parameter spaces. It further introduces a gradient-based filtering step to mask distractors during inference. It uses a novel adaptive importance sampling approach that efficiently identifies posterior samples that explain *all* available iid observations. We offer an efficient `JAX`-based implementation that leverages key auto-differentiation and vectorization features to accelerate the inference. Simulations performed with novel experimental settings as well as the comprehensive SBI benchmark show that R2OMC achieves improved accuracy (close to the optimal 0.5 `C2ST`) compared to existing methods, while using a smaller number of simulator calls and a fraction of the computational time.

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Checklist

The checklist follows the references. For each question, choose your answer from the three possible options: Yes, No, Not Applicable. You are encouraged to include a justification to your answer, either by referencing the appropriate section of your paper or providing a brief inline description (1-2 sentences). Please do not modify the questions. Note that the Checklist section does not count towards the page limit. Not including the checklist in the first submission won't result in desk rejection, although in such case we will ask you to upload it during the author response period and include it in camera ready (if accepted).

In your paper, please delete this instructions block and only keep the Checklist section heading above along with the questions/answers below.

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [No]
2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Not Applicable]
   (b) Complete proofs of all theoretical results. [Not Applicable]
   (c) Clear explanations of any assumptions. [Not Applicable]
3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [No] Not now, but code will be made available if accepted.
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
   (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes]
   (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
   (a) Citations of the creator If your work uses existing assets. [Yes]
   (b) The license information of the assets, if applicable. [Yes]
   (c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable]
   (d) Information about consent from data providers/curators. [Not Applicable]
   (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
   (a) The full text of instructions given to participants and screenshots. [Not Applicable]
   (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
   (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

A Methods

Throughout the paper we run experiments based on four methods: R2OMC (our proposal), NAIVE-ROMC (a straightforward use of the ROMC framework by Ikonomov and Gutmann, 2020), Neural Posterior Estimation (NPE) (Greenberg et al., 2019), Neural Likelihood Estimation (NLE) (Papamakarios et al., 2019).

R2OMC. A detailed description of the method can be found in Algorithm 1.

In Step 2 of Algorithm 1, i.e., computing the uninformative dimensions, we typically use around 50 samples from the distribution $p(u)$ and another 50 samples from $p(\boldsymbol{\theta})$, with the parameter τ set to machine precision, i.e., $\tau \approx 0$. Similar results can be obtained with fewer samples, such as 10 or 20.

For Step 3, i.e., computing the optimal points $\boldsymbol{\theta}^*_{i,n}$, we use the Adam optimizer, with a learning rate between 0.01 and 0.2, depending on the specific problem. Our default choice is 0.1. The number of optimization steps varies between 100 and 400, with 50 as the default. The distance function $d(\boldsymbol{\theta})$ is defined as the squared Euclidean distance between the simulator output and the observation, i.e., $d(\boldsymbol{\theta}) = \|\mathbf{y} - \mathbf{y}^o_n\|_2^2$. The number of seeds S used ranges from 500 to 5000, with 1000 as the default. The number of seeds corresponds to the simulation budget; for example, if a budget of 10,000 runs is available, we use (at most) 10,000 seeds. Typically, we select 80% of the seeds based on the best $d^n_i(\boldsymbol{\theta}^*_{i,n})$. However, the percentage of accepted seeds can vary between 50% and 100%, depending on the absolute values of $d^n_i(\boldsymbol{\theta}^*_{i,n})$ for $i = 1, \ldots, S$.

For Step 5, i.e., constructing the hyperboxes, we use Algorithm 2. We normally set ϵ to be twice the distance of the 'worst' accepted seed, i.e., $\epsilon = 2 \max_i d^n_i(\boldsymbol{\theta}^*_{i,n})$, where i is an index over the accepted seeds. Alternatively, ϵ can be set to a fixed value such as 0.1. The typical settings for step size, number of steps, and refinements are $\eta = 0.1$, $L = 100$, and $R = 1$, respectively.

For Step 7, i.e., sampling from the proposal distribution, we generate between 1000 and 100,000 candidate samples, with the default being 5000. From these candidates, we select 1000 via weighted sampling. For computing the weights, $\{w_p\}_{p=1}^P$, we set ϵ to a value about twice the distance of the 'worst' accepted seed, i.e., $\epsilon = 2 \max_{i,n} d^n_i(\boldsymbol{\theta}^*_{i,n})$. However, this can vary depending on the problem, and by testing different values.

Finally, we keep 1000 posterior samples, which are used to compute the `C2ST` score using another 1000 samples drawn from the true posterior.

Algorithm 2 Hyperbox Computation

Input: Optimal point $\boldsymbol{\theta}^*$, distance function $d(\boldsymbol{\theta})$, Jacobian $\mathbf{J}$ at $\boldsymbol{\theta}^*$

Parameters: Step size η, number of refinements R, number of steps L, distance threshold ϵ

Compute eigenvectors $\mathbf{v}_d$ of $\mathbf{J}^T\mathbf{J}$ $(d = 1, \ldots, D)$
for $d = 1$ to D **do**
  Initialize endpoint $\tilde{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}^*$
  Set refinement counter $r \leftarrow 0$
  repeat
    Set step counter $l \leftarrow 0$
    repeat
      Increment step counter $l \leftarrow l + 1$
      Update $\tilde{\boldsymbol{\theta}} \leftarrow \tilde{\boldsymbol{\theta}} + \eta \mathbf{v}_d$
    until $d(\boldsymbol{\theta}) > \epsilon$ or $l = L$
    Move back one step $\tilde{\boldsymbol{\theta}} \leftarrow \tilde{\boldsymbol{\theta}} - \eta \mathbf{v}_d$
    Halve the step size $\eta \leftarrow \eta/2$
    Increment refinement counter $r \leftarrow r + 1$
  until $r = R$
  Set $\tilde{\boldsymbol{\theta}}$ as the endpoint
  Repeat steps 3-15 for $\mathbf{v}_d = -\mathbf{v}_d$ and set $\tilde{\boldsymbol{\theta}}$ as the negative endpoint
end for
Define a uniform distribution q over the hyperbox
return q

NAIVE-ROMC. We use the Algorithm presented as ROMC in Ikonomov and Gutmann (2020). The characterization NAIVE indicates the use of a naive distance function in case of multiple observations. As described in the main paper, in case of N observations, we concatenate them in a single vector $\mathbf{y}^o = (\mathbf{y}^{o,(1)}, \dots, \mathbf{y}^{o,(N)})$, then, run the simulator N times, concatenate the ouputs, $\mathbf{y} = (\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(N)})$ and set the distance to be the squared Euclidean distance between the two vectors, $d(\boldsymbol{\theta}) = ||\mathbf{y} - \mathbf{y}^o||_2^2$. Although using this approach for defining $d(\boldsymbol{\theta})$ has disadvantages, that is why we name it 'NAIVE', there is no other (simple) approach that leads to a better posterior approximation, under multiple observations.

NAIVE-ROMC is practically the same as R2OMC (Algorithm 1 of the main text), if (a) we omit Step 2, i.e., computing the uninformative dimensions and (b) set $N = 1$. This is why NAIVE-ROMC and R2OMC have a similar `C2ST` score in problems without distractors and with a single observation. In case of multiple observations, we use as $\mathbf{y}$ and $\mathbf{y}^o$, the concatenated vectors as described above. For NAIVE-ROMC, we use a `JAX`-based implementation that is similar to the one we use for R2OMC. The reason for not using its existing implementation (Gkolemis et al., 2023) of the ELFI package (Lintusaari et al., 2018) is that it does not use gradients, so it cannot scale to high dimensions. The hyperparameters used in NAIVE-ROMC are similar to R2OMC, as described above.

NPE As Neural Posterior Estimation (NPE) we use the one denoted as as SNPE_C in the SBI benchmark (Lueckmann et al., 2021), which is a method proposed by Greenberg et al. (2019). SNPE_C estimates the posterior by training a neural network $F(\mathbf{y}, \phi)$ to approximate $p(\boldsymbol{\theta}|\mathbf{y})$ using a density estimator $q_F(\mathbf{y}, \phi)(\boldsymbol{\theta})$. For further details, see (Greenberg et al., 2019).

We use the `Pytorch` implementation provided by the SBI package, using mostly the default parameters. As density estimator we use the a neural network with 50 hidden units, `sbi.utils.get_nn_models.posterior_nn()` with parameters `model='nsf'`, `hidden_features=50`, `z_score_x='independent'`, and `z_score_x='independent'`. During training, we use 10 atoms, a batch size of 1000 (or the whole dataset) and a maximum number of 500 epochs.

We use a simulation budget of $10,000$ simulator executions, which means that we train the density estimator on $10,000$ samples. We normally use 2 training rounds, each with 5000 training samples: the first set of 5000 is drawn from the prior, while the second set is sampled from the approximate posterior learned after the first round. Occasionally, if the initial 5000 samples do not yield a reliable approximation and conditioning on $\mathbf{y}^o$ leads to instabilities, we use a single round of 10000 samples.

In cases with multiple observations, since NPE cannot handle them separately, we follow the same approach as in NAIVE-ROMC(see above); we concatenate the observations and the outputs to one vector each.

NLE As NLE we use the method proposed by Papamakarios et al. (2019). NLE approximates the likelihood $p(\mathbf{y}^o|\theta)$ using neural density estimation. The process involves generating a dataset $D = \{(\boldsymbol{\theta}_k, \mathbf{y}_k)\}_{k=1}^K$ by sampling $\boldsymbol{\theta}_k \sim p(\boldsymbol{\theta})$ and simulating $\mathbf{y}_k \sim p(\mathbf{y}|\boldsymbol{\theta}_k)$. A conditional neural density estimator $q_\psi(\mathbf{y}|\boldsymbol{\theta})$ is then trained to model the likelihood.

Once trained, samples from the posterior $\hat{p}(\boldsymbol{\theta}|\mathbf{y}^o) \propto q_\psi(\mathbf{y}^o|\boldsymbol{\theta})p(\boldsymbol{\theta})$ are obtained using MCMC. For density estimation, we use the default SBI settings, i.e., a Masked Autoregressive Flow (MAF) with five flow transforms and 50 hidden units per block. For MCMC, 250 warm-up steps and posterior samples are obtained with thinning.

We use a simulation budget of $10,000$ simulator executions, which means that we train the density estimator on $10,000$ samples. We always use a single training round. NLE can handle multiple observations; we simply pass a `numpy` array of shape: `(N,D)`, where N is the number of observations and D their dimensionality. Internally, NLE evaluates each observation using the neural density estimator $q_\psi(\mathbf{y}|\boldsymbol{\theta})$ and multiplies them to estimate the joint likelihood and then uses MCMC to sample from the posterior.

B Discussion about simulation budget

TODO

C Introductory Example

Notation. We use $\mathcal{N}(\mathbf{y}; \boldsymbol{\theta}, \Sigma)$ to denote the probability density function (pdf) of a Gaussian random variable $\mathbf{y}$, while $\mathcal{N}(\boldsymbol{\theta}, \Sigma)$ denotes the random variable itself. Typically, $\mathcal{N}(\mathbf{y}; \boldsymbol{\theta}, \Sigma)$ is used when deriving expressions, whereas $\mathcal{N}(\boldsymbol{\theta}, \Sigma)$ is used to indicate sampling, such as in $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\theta}, \Sigma)$. We use $\mathcal{U}(\mathbf{a}, \mathbf{b})$ to denote a multivariate random variable that is uniformly distributed on the hypercube defined by vector $\mathbf{a}$ with the lower boundaries and the vector $\mathbf{b}$ with the upper boundaries. The dimensions are typically clear from the context if not explicitly stated. Bold indicates a vector, e.g., $\boldsymbol{\theta} \in \mathbb{R}^D$.

Setup. We provide further details about the introductory example shown in Figure 1. The introductory example consists of five different versions: (1) Simple, (2) High-dimensional, (3) Distractor, (4) Multiple observations, and (5) Combined.

The simulator of the Simple (1), High-dimensional (2) and Multiple observations (4) versions is:

$$\mathbf{y} \sim \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \mathbf{I})\right) \tag{15}$$

At the Distractor (3) and the Combined (5) versions, we add $D_{\text{dist}} = 10$ distractor dimensions to the output, so the simulator becomes:

$$(\mathbf{y}^{(1)}, \mathbf{y}^{(2)}), \quad \mathbf{y}^{(1)} \sim \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \mathbf{I})\right), \quad \mathbf{y}^{(2)} \sim \mathcal{N}(\boldsymbol{\mu}, 100\mathbf{I}), \tag{16}$$

where $\boldsymbol{\mu} \sim \mathcal{U}(-\mathbf{10}, \mathbf{10})$. In all cases the prior is given by $p(\boldsymbol{\theta}) = \mathcal{U}(-\mathbf{15}, \mathbf{15})$.

C.1 Simple Case

The simulator is the one in (15) with $D = 5$ and the observation is $\mathbf{y}^o = (5, \dots, 5) + \epsilon$ where $\mathbf{y}^o \in \mathbb{R}^5$. The ground truth posterior for $\mathbf{y}^o = (5, \dots, 5)$ is given by:

$$\begin{aligned}
p(\boldsymbol{\theta}|\mathbf{y}^o) &\propto p(\boldsymbol{\theta}) p(\mathbf{y}^o|\boldsymbol{\theta}) && (17)\\
&\propto p(\boldsymbol{\theta})(\mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(\mathbf{y}^o; -\boldsymbol{\theta}, \mathbf{I})) && (18)\\
&\propto p(\boldsymbol{\theta})(\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}^o, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{y}^o, \mathbf{I})) && (19)\\
&\propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^5, \\ 0, & \text{otherwise.} \end{cases} && (20)
\end{aligned}$$

From (18) to (19), we apply the property: $\mathcal{N}(\mathbf{y}; -\boldsymbol{\theta}, \Sigma) = \mathcal{N}(\boldsymbol{\theta}; -\mathbf{y}, \Sigma)$.

C.2 High-dimensional Case

The simulator is the one in (15) with $D = 20$ and the observation is $\mathbf{y}^o = (5, \dots, 5) + \epsilon$ where $\mathbf{y}^o \in \mathbb{R}^{20}$. The derivation of (20) holds irrespectively of the dimensionality, so the ground truth posterior for $\mathbf{y}^o = (5, \dots, 5)$ is:

$$p(\boldsymbol{\theta}|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^{20}, \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

C.3 Distractor Case

The simulator is the one in (16) with $D = 5$ and $D_{\text{dist}} = 10$, leading to an output dimensionality of $D_y + D_{\text{dist}} = 15$. The observation is $\mathbf{y}^o = (\mathbf{y}^{o,(1)}, \mathbf{y}^{o,(2)})$, where $\mathbf{y}^{o,(1)} = (5, \dots, 5) + \epsilon$ and $\mathbf{y}^{o,(2)} = (0, \dots, 0) + \epsilon$. Below, we show

that the ground truth posterior is equivalent to the simple case (20), as the distractor dimensions do not affect the posterior:

$$
\begin{aligned}
p(\boldsymbol{\theta}|\mathbf{y}^o) &\propto p(\boldsymbol{\theta})p(\mathbf{y}^o|\boldsymbol{\theta}) && (22)\\
&\propto p(\boldsymbol{\theta})p((\mathbf{y}^{o,(1)}, \mathbf{y}^{o,(2)})|\boldsymbol{\theta}) && (23)\\
&\propto p(\boldsymbol{\theta})p(\mathbf{y}^{o,(1)}|\boldsymbol{\theta})p(\mathbf{y}^{o,(2)}|\boldsymbol{\theta}) && (24)\\
&\propto p(\boldsymbol{\theta})p(\mathbf{y}^{o,(1)}|\boldsymbol{\theta})p(\mathbf{y}^{o,(2)}) && (25)\\
&\propto p(\boldsymbol{\theta})p(\mathbf{y}^{o,(1)}|\boldsymbol{\theta}) && (26)\\
&\propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^5, \\ 0, & \text{otherwise.} \end{cases} && (27)
\end{aligned}
$$

From (23) to (24), we use that $\mathbf{y}^{o,(1)}$ is independent of $\mathbf{y}^{o,(2)}$ and from (24) to (25), that $\mathbf{y}^{o,(2)}$ is independent of $\boldsymbol{\theta}$.

C.4 Multiple Observations Case

The simulator is the one in (15) with $D = 5$ and the set of $N = 4$ iid observations is $\{\mathbf{y}_n^o\}_n^N$ where $\mathbf{y}_n^o = (5, \dots, 5) + \epsilon$ for $n = 1, 2$ and $\mathbf{y}_n^o = (-5, \dots, -5) + \epsilon$ for $n = 3, 4$. We will derive the ground truth posterior for the general case of N observations, where $\mathbf{y}_n^o = (5, \dots, 5) + \epsilon$ for $n \in \{1, \dots, N/2\}$ and $\mathbf{y}_n^o = (-5, \dots, -5) + \epsilon$ for $n \in \{N/2+1, \dots, N\}$.

$$
\begin{aligned}
p(\boldsymbol{\theta}|\{\mathbf{y}_k^o\}) &\propto p(\boldsymbol{\theta}) \prod_n^N p(\mathbf{y}_n^o|\boldsymbol{\theta}) && (28)\\
&\propto p(\boldsymbol{\theta}) \prod_{n=1}^{N/2} p(\mathbf{y}_n^o|\boldsymbol{\theta}) \prod_{n=N/2+1}^{N} p(\mathbf{y}_n^o|\boldsymbol{\theta}) && (29)\\
&\propto p(\boldsymbol{\theta}) \prod_{n=1}^{N/2} (\mathcal{N}(\mathbf{5}; \boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(-\mathbf{5}; \boldsymbol{\theta}, \mathbf{I})) \prod_{n=N/2+1}^{N} (\mathcal{N}(-\mathbf{5}; \boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(\mathbf{5}; \boldsymbol{\theta}, \mathbf{I})) && (30)\\
&\propto p(\boldsymbol{\theta}) \prod_{n=1}^{N} (\mathcal{N}(\mathbf{5}; \boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(-\mathbf{5}; \boldsymbol{\theta}, \mathbf{I})) && (31)\\
&\propto p(\boldsymbol{\theta}) \prod_{n=1}^{N} (\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})) && (32)\\
&\propto\!\!\!\!\sim\, p(\boldsymbol{\theta})(\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}))^N + (\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}))^N && (33)\\
&\propto\!\!\!\!\sim\, p(\boldsymbol{\theta})(\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \tfrac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \tfrac{1}{N}\mathbf{I})) && (34)\\
&\propto\!\!\!\!\sim \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \frac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \frac{1}{N}\mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^5, \\ 0, & \text{otherwise.} \end{cases} && (35)
\end{aligned}
$$

From (29) to (30), we simply plug in the corresponding observations. From (31) to (32), we apply the property: $\mathcal{N}(\mathbf{y}; -\boldsymbol{\theta}, \Sigma) = \mathcal{N}(\boldsymbol{\theta}; -\mathbf{y}, \Sigma)$. From (32) to (33), we use that the terms that include the product $\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I})\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})$ have a much smaller magnitude than the two terms $\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I})^N$ and $\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})^N$. Therefore, we keep only the latter in (33). From (33) to (34), we use the property: $\mathcal{N}(\boldsymbol{\theta}; \boldsymbol{\alpha}, \Sigma)^N = \mathcal{N}(\boldsymbol{\theta}; \boldsymbol{\alpha}, \frac{1}{N}\Sigma)$.

C.5 Combined Case

The simulator is the one in (16) with $D = 20$ and $D_{\text{dist}} = 10$, leading to an output dimensionality of $D_y + D_{\text{dist}} = 30$. The set of $N = 4$ iid observations is $\{\mathbf{y}_n^o\}_n^N$ where $\mathbf{y}_n^o = (\mathbf{y}^{o,(1)}, \mathbf{y}^{o,(2)})$, with $\mathbf{y}^{o,(1)}$ equal $(5, \dots, 5) + \epsilon$ for $n \in \{1, \dots, N/2\}$ and $(-5, \dots, -5) + \epsilon$ for $n \in \{N/2 + 1, \dots, N\}$ and $\mathbf{y}^{o,(2)}$ equal $(0, \dots, 0) + \epsilon$ for all n.

Using the properties of the previous cases we obtain the ground truth posterior, which is the same as in the

multiple observations case (35) for $D = 20$:

$$p(\boldsymbol{\theta}|\{\mathbf{y}_k^o\}) \propto p(\boldsymbol{\theta}) \prod_n^N p(\mathbf{y}_n^o|\boldsymbol{\theta}) \tag{36}$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N p((\mathbf{y}^{o,(1)}, \mathbf{y}^{o,(2)})|\boldsymbol{\theta}) \tag{37}$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N p(\mathbf{y}^{o,(1)}|\boldsymbol{\theta}) p(\mathbf{y}^{o,(2)}|\boldsymbol{\theta}) \tag{38}$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N p(\mathbf{y}^{o,(1)}|\boldsymbol{\theta}) p(\mathbf{y}^{o,(2)}) \tag{39}$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N p(\mathbf{y}^{o,(1)}|\boldsymbol{\theta}) \tag{40}$$

From (37) to (38), we use that the observations are independent and from (38) to (39) that $\mathbf{y}^{o,(2)}$ is independent of $\boldsymbol{\theta}$. Using the derivation of the multiple observations case (35) gives

$$p(\boldsymbol{\theta}|\{\mathbf{y}_k^o\}) \propto\!\!\!\propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \frac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \frac{1}{N}\mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^{20}, \\ 0, & \text{otherwise.} \end{cases} \tag{41}$$

C.6 Results

Setup. For NAIVE-ROMCand R2OMC, a maximum of $S = 5000$ seeds was used, corresponding to approximately 5000 independent simulator calls, which is (approximately) a simulation budget of 5000. NLE and NPE were trained with a simulation budget of 10000 samples. For NPE, we used $R = 2$ rounds with 5000 samples each, while for NLE, we used a single round of 10000 samples. The default settings from the SBI library Lueckmann et al. (2021) were utilized for both, with a training batch size of 1000 and a maximum of 300 training epochs.

Conclusions. The results of the introductory example are presented in Figure 1 in the main text. In the simple scenario, all methods provide a reasonably accurate approximation of the ground truth posterior. In the high-dimensional case, both NPE and NLE encounter difficulties, indicating that high-dimensional problems are challenging for these methods even when the simulator is relatively simple. In contrast, NAIVE-ROMC and R2OMC yield a good approximation of the ground truth posterior, demonstrating their robustness in high-dimensional settings. Note that in the simple scenario, NAIVE-ROMC and R2OMC are equivalent, since there are no distractors and only one observation. In the distractors scenario, NPE struggles, NLE is more robust than NPE, but still faces challenges in providing an accurate approximation, while NAIVE-ROMC struggles. R2OMC provides a good approximation of the ground truth posterior. In the multiple observations scenario, only NLE and R2OMC succeed in generating a reliable approximation of the ground truth posterior. Finally, in the combined scenario, only R2OMC delivers a satisfactory approximation of the ground truth posterior.

D Additional information on the experiments

We here provide some additional information on the experiments presented in the main text.

D.1 MoG Benchmark - High Dimensional Simulators

D.1.1 Base simulator - without distractors

Prior $\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{15}, \mathbf{15})$

Simulator $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$, with $\sigma^2 = 1.5^2$

Dimensionality $\boldsymbol{\theta} \in \mathbb{R}^D$, $\mathbf{y} \in \mathbb{R}^D$

Observation $\mathbf{y}^o = (5, \dots, 5) + \epsilon$ where $\mathbf{y}^o \in \mathbb{R}^D$

Posterior $p(\boldsymbol{\theta}|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \sigma^2 \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases}$

Proof of the posterior:

$$
\begin{aligned}
p(\boldsymbol{\theta}|\mathbf{y}^o) &\propto p(\boldsymbol{\theta}) p(\mathbf{y}^o|\boldsymbol{\theta}) && (42)\\
&\propto p(\boldsymbol{\theta}) \mathcal{N}(\mathbf{y}^o; \boldsymbol{\theta}, \sigma^2 \mathbf{I}) && (43)\\
&\propto p(\boldsymbol{\theta}) \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}^o, \sigma^2 \mathbf{I}) && (44)\\
&\propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \sigma^2 \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases} && (45)
\end{aligned}
$$

From (43) to (44), we apply the property: $\mathcal{N}(\mathbf{y}; -\boldsymbol{\theta}, \Sigma) = \mathcal{N}(\boldsymbol{\theta}; -\mathbf{y}, \Sigma)$.

D.1.2 Base simulator - with distractors

Prior $\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{15}, \mathbf{15})$

Simulator $\mathbf{y} = (\mathbf{y^{(1)}}, \mathbf{y^{(2)}})$,
$\mathbf{y^{(1)}} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$, $\sigma^2 = 1.5^2$,
$\mathbf{y^{(2)}} \sim \mathcal{N}(\boldsymbol{\mu}, 100\mathbf{I})$, $\boldsymbol{\mu} \sim \mathcal{U}(-\mathbf{10}, \mathbf{10})$

Dimensionality $\boldsymbol{\theta} \in \mathbb{R}^D, \mathbf{y} \in \mathbb{R}^{D+100}$, $\mathbf{y}^{(1)} \in \mathbb{R}^D$, $\mathbf{y}^{(2)} \in \mathbb{R}^{100}$

Observation $\mathbf{y}^o = (\mathbf{y^{o,(1)}}, \mathbf{y^{o,(2)}})$, where $\mathbf{y^{o,(1)}} = (5, \dots, 5) + \epsilon$ and $\mathbf{y^{o,(2)}} = (0, \dots, 0) + \epsilon$

Posterior $p(\boldsymbol{\theta}|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \sigma^2 \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases}$

The ground truth posterior is the same as in (45), as the distractors do not affect the posterior, see (27).

D.1.3 MoG simulator - without distractors

Prior $\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{15}, \mathbf{15})$

Simulator $\mathbf{y} \sim \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta}, \sigma_1^2 \mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \sigma_2^2 \mathbf{I})\right)$, $\sigma_1^2 = 0.1$, $\sigma_2^2 = 1$

Dimensionality $\boldsymbol{\theta} \in \mathbb{R}^D$, $\mathbf{y} \in \mathbb{R}^D$

Observation $\mathbf{y}^o = (5, \dots, 5) + \epsilon$ where $\mathbf{y}^o \in \mathbb{R}^D$

Posterior $p(\boldsymbol{\theta}|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \sigma_1^2 \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \sigma_2^2 \mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases}$

The MoG simulator is the same as in the introductory example, so the proof is equivalent to Section C.

D.1.4 MoG simulator - with distractors

Prior	$\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{15}, \mathbf{15})$
Simulator	$\mathbf{y} = (\mathbf{y^{(1)}}, \mathbf{y^{(2)}})$,
	$\mathbf{y^{(1)}} \sim \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta}, \sigma_1^2\mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \sigma_2^2\mathbf{I})\right)$, $\sigma_1^2 = 0.1$, $\sigma_2^2 = 1$,
	$\mathbf{y^{(2)}} \sim \mathcal{N}(\boldsymbol{\mu}, 100\mathbf{I})$, $\boldsymbol{\mu} \sim \mathcal{U}(-\mathbf{10}, \mathbf{10})$
Dimensionality	$\boldsymbol{\theta} \in \mathbb{R}^D$, $\mathbf{y} \in \mathbb{R}^{D+100}$, $\mathbf{y}^{(1)} \in \mathbb{R}^D$, $\mathbf{y}^{(2)} \in \mathbb{R}^{100}$
Observation	$\mathbf{y}^o = (\mathbf{y^{o,(1)}}, \mathbf{y^{o,(2)}})$, where $\mathbf{y^{o,(1)}} = (5, \dots, 5) + \epsilon$ and $\mathbf{y^{o,(2)}} = (0, \dots, 0) + \epsilon$
Posterior	$p(\boldsymbol{\theta}\|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \sigma_1^2\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \sigma_2^2\mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases}$

The MoG simulator is the same as in the introductory example, so the proof is equivalent to Section C.

D.2 MoG Benchmark - Multiple Observations

D.2.1 Base simulator

Prior	$\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{15}, \mathbf{15})$
Simulator	$\mathbf{y} \sim \mathcal{N}(\mathbf{y}; \boldsymbol{\theta}, \sigma^2\mathbf{I})$, $\sigma^2 = 1$
Dimensionality	$\boldsymbol{\theta} \in \mathbb{R}^D$, $\mathbf{y} \in \mathbb{R}^D$
Observations	$\{\mathbf{y}_n^o\}_n^N$ where $\mathbf{y}_n^o = (5, \dots, 5) + \epsilon$ for $n \in \{1, \dots, 4\}$
Posterior	$p(\boldsymbol{\theta}\|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \frac{1}{N}\sigma^2\mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases}$

Proof of the posterior:

$$p(\boldsymbol{\theta}|\{\mathbf{y}_n^o\}) \propto p(\boldsymbol{\theta}) \prod_n^N p(\mathbf{y}_n^o|\boldsymbol{\theta}) \tag{46}$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N \mathcal{N}(\mathbf{y}_n^o; \boldsymbol{\theta}, \sigma^2\mathbf{I}) \tag{47}$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \sigma^2\mathbf{I}) \tag{48}$$

$$\propto p(\boldsymbol{\theta}) \prod_{n=1}^N \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \sigma^2\mathbf{I}) \tag{49}$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \frac{1}{N}\sigma^2\mathbf{I}) \tag{50}$$

$$\propto \begin{cases} p(\boldsymbol{\theta})\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \frac{1}{N}\sigma^2\mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases} \tag{51}$$

In Figure 8, we present additional results for the base simulator with multiple observations.

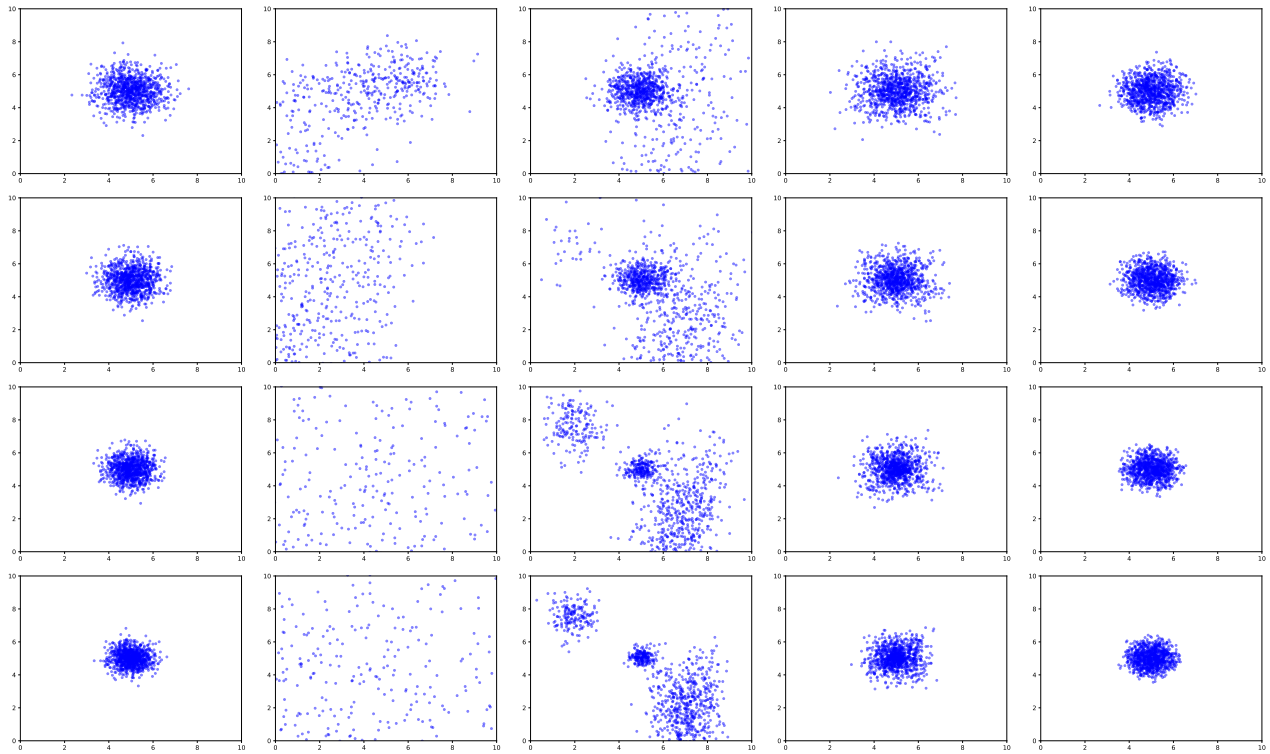


Figure 8: Multi observations Base: Left to right: ground truth, NPE, NLE, Naive-ROMC, R2OMC. Top to bottom: $N = 3$, $N = 5$, $N = 10$, $N = 10$

D.2.2 MoG simulator

Prior $\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{15}, \mathbf{15})$

Simulator $\mathbf{y} \sim \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta}, \sigma_1^2\mathbf{I}) + \mathcal{N}(-\boldsymbol{\theta}, \sigma_2^2\mathbf{I})\right)$, with $\sigma_1 = \sigma_2 = 1$

Dimensionality $\boldsymbol{\theta} \in \mathbb{R}^D$, $\mathbf{y} \in \mathbb{R}^D$

Observations $\{\mathbf{y}_n^o\}_n^N$ where:

$\mathbf{y}_n^o = (5, \dots, 5) + \epsilon$ for $n \in \{1, \dots, \frac{N}{2}\}$,

$\mathbf{y}_n^o = (-5, \dots, -5) + \epsilon$ for $n \in \{\frac{N}{2} + 1, \dots, N\}$

Posterior $p(\boldsymbol{\theta}|\mathbf{y}^o) \propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \frac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \frac{1}{N}\mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases}$

Proof of the posterior:

$$p(\boldsymbol{\theta}|\{\mathbf{y}_n^o\}) \propto\!\!\!\propto \begin{cases} \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \frac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \frac{1}{N}\mathbf{I}), & \text{if } \boldsymbol{\theta} \in [-15, 15]^D, \\ 0, & \text{otherwise.} \end{cases} \tag{52}$$

The proof is similar to the one in Eq. (35).

In Figure 9, we present the results for the MoG simulator with multiple observations.

D.3 SBI - Mixtures of Gaussians (T.7)

We here provide additional information on example T.7 of the SBI benchmark Lueckmann et al. (2021). It tests inference of the mean of a mixture of two 2-D Gaussians, one with much broader covariance than the other

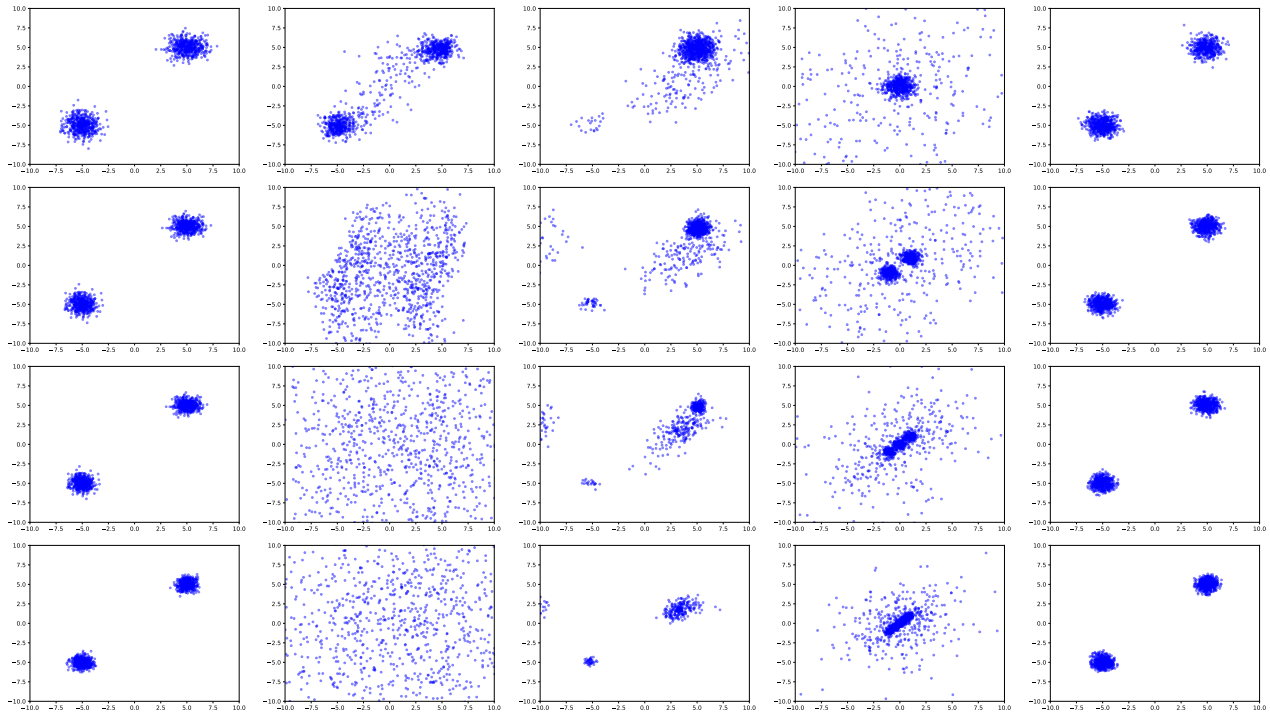
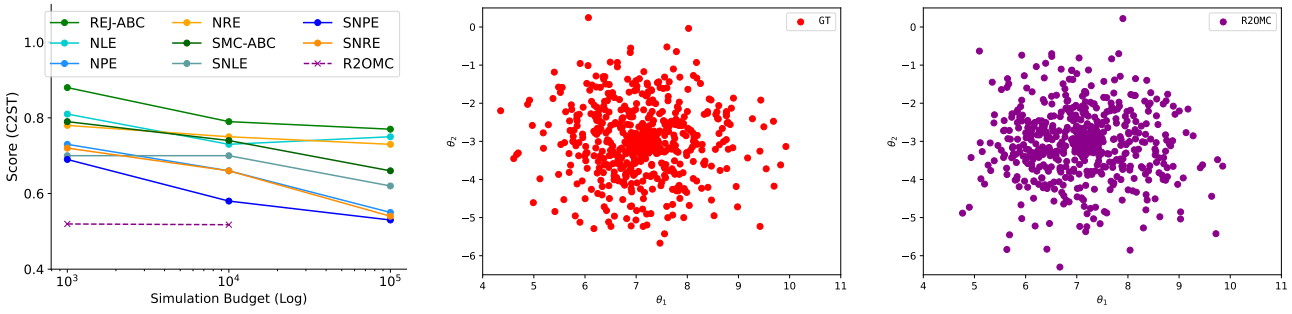


Figure 9: Multi observations MoG: Left to right: ground truth, NPE, NLE, Naive-ROMC, R2OMC. Top to bottom: $N = 3$, $N = 5$, $N = 10$, $N = 10$

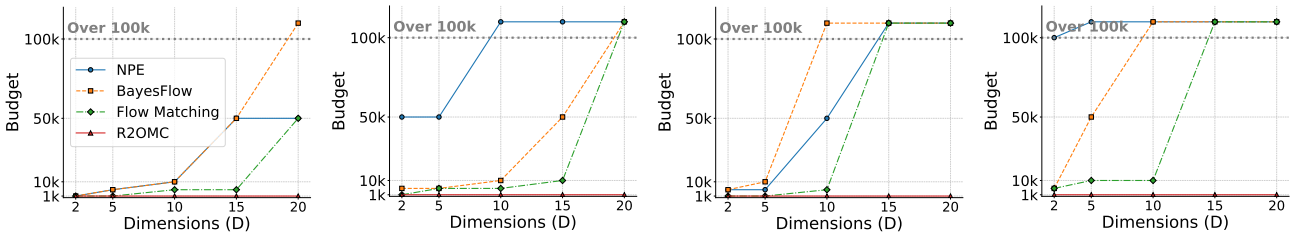
Prior	$\boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{10}, \mathbf{10})$
Simulator	$\mathbf{y} \sim 0.5(\mathcal{N}(\boldsymbol{\theta}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}, \sigma^2\mathbf{I}))$, with $\sigma^2 = 0.01$
Dimensionality	$\boldsymbol{\theta} \in \mathbb{R}^2$, $\mathbf{y} \in \mathbb{R}^2$

Results. We evaluate R2OMC using a simulation budget of $S = 1000$ and $S = 10000$ seeds. The `C2ST` score is computed by comparing 1000 samples from the true posterior (we randomly select 1000 from the 10000 that are provided by the SBI benchmark) with 1000 samples generated from the R2OMC approximate posterior. The results are illustrated in Figure 10.



(a) `C2ST` score (b) Ground truth samples (c) Samples by R2OMC

Figure 10: T.7: Gaussian Mixture. From left to right: `C2ST` score, ground truth samples, and samples by R2OMC.



(a) $S_{\texttt{base}}$ (b) $S^{\texttt{dist}}_{\texttt{base}}$ (c) $S_{\texttt{MoG}}$ (d) $S^{\texttt{dist}}_{\texttt{MoG}}$

Figure 11: MoG benchmark: Success Frontier plots showing the lowest budget required to reach a C2ST score ≤ 0.75 for varying D.

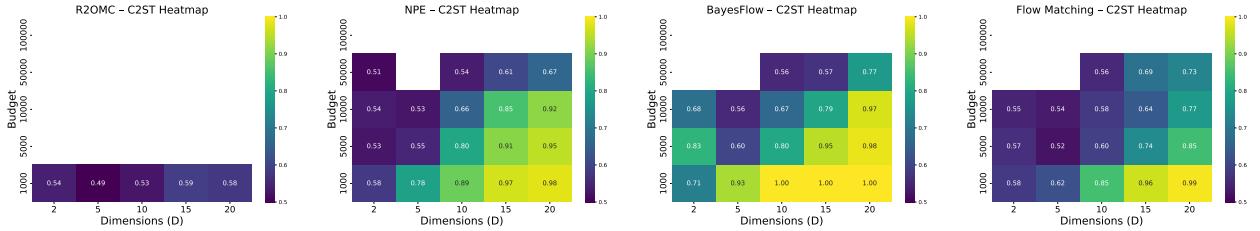


Figure 12: MoG Benchmark - Base simulator: C2ST heatmaps for R2OMC, NPE, BayesFlow, and Flow Matching.

D.4 MoG Benchmark

D.5 SBI - SLCP (T.3) and SLCP Distractors (T.4)

D.6 SBI - Two Moons (T.8)

We here provide additional information on example T.8 of the SBI benchmark Lueckmann et al. (2021). It tests inference when the posterior exhibits both global (bimodality) and local (crescent shape) structure to illustrate how algorithms deal with multimodality:

Prior $\quad \boldsymbol{\theta} \sim \mathcal{U}(-\mathbf{1}, \mathbf{1})$

Simulator $\quad \mathbf{y} \sim \begin{bmatrix} r\cos(\alpha) + 0.25 \\ r\sin(\alpha) \end{bmatrix} + \begin{bmatrix} |\theta_1 + \theta_2|/\sqrt{2} \\ (-\theta_1 + \theta_2)/\sqrt{2} \end{bmatrix}$, where $r \sim \mathcal{N}(0.1, 0.01^2)$ and $\alpha \sim \mathcal{U}(-\pi/2, \pi/2)$

Dimensionality $\quad \boldsymbol{\theta} \in \mathbb{R}^2$, $\mathbf{y} \in \mathbb{R}^2$

Results. We evaluate R2OMC using a simulation budget of $S = 1000$ and $S = 10000$ seeds. The `C2ST` score is computed by comparing 1000 samples from the true posterior (we randomly select 1000 from the 10000 that are provided by the SBI benchmark) with 1000 samples generated from the R2OMC approximate posterior. The results are illustrated in Figure 10. We observe that R2OMC is able to capture the bimodal structure of the posterior, perfectly replicating the ground truth samples.

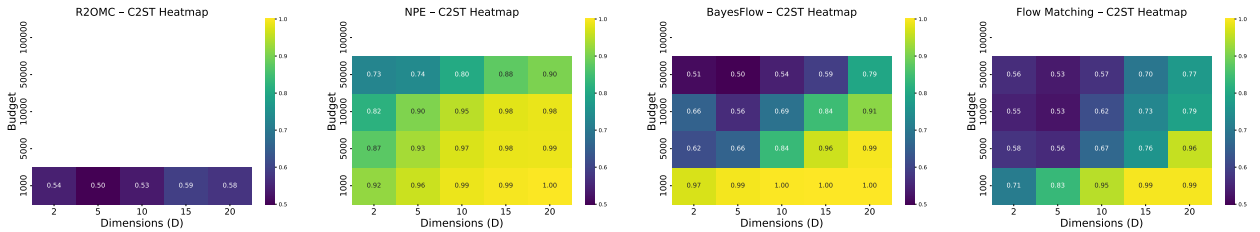


Figure 13: MoG Benchmark - Base simulator with distractors: C2ST heatmaps for R2OMC, NPE, BayesFlow, and Flow Matching.

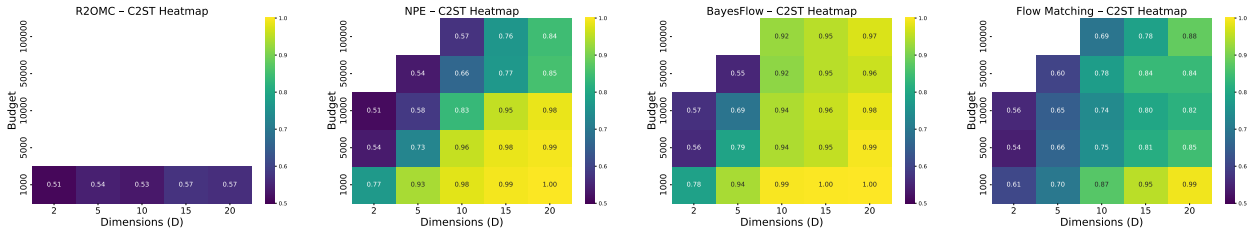


Figure 14: MoG Benchmark - MoG simulator: C2ST heatmaps for R2OMC, NPE, BayesFlow, and Flow Matching.

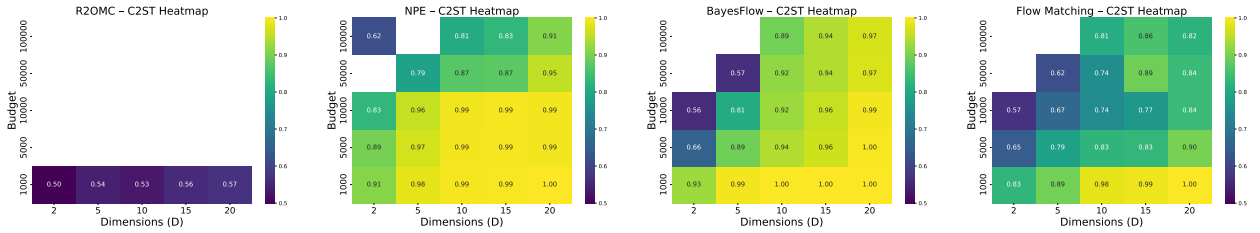


Figure 15: MoG Benchmark - MoG simulator with distractors: C2ST heatmaps for R2OMC, NPE, BayesFlow, and Flow Matching.

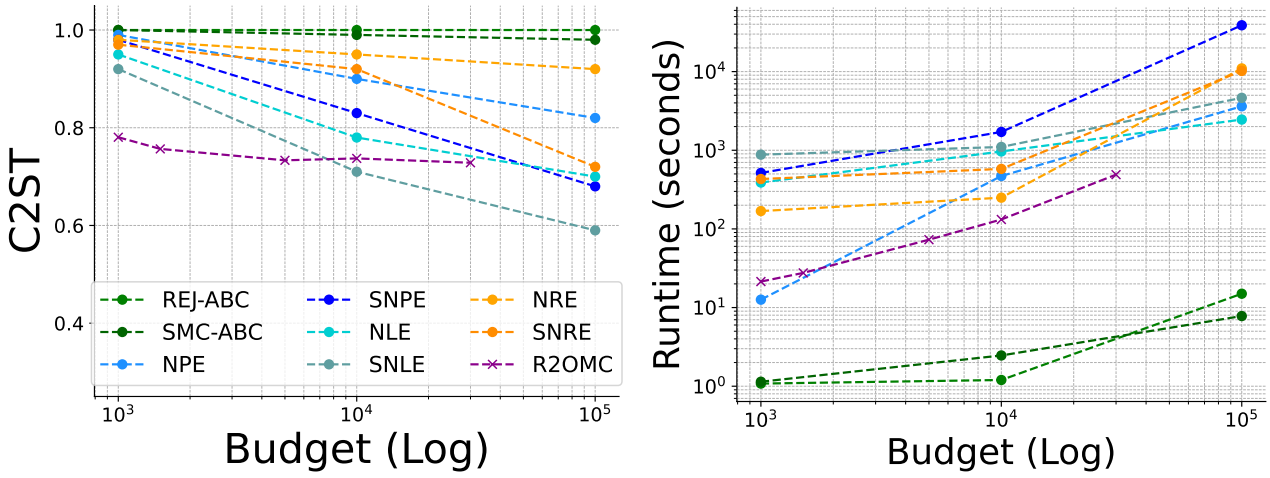


Figure 16: T.3: SLCP. From left to right: `C2ST` vs. budget, and runtime vs. budget.

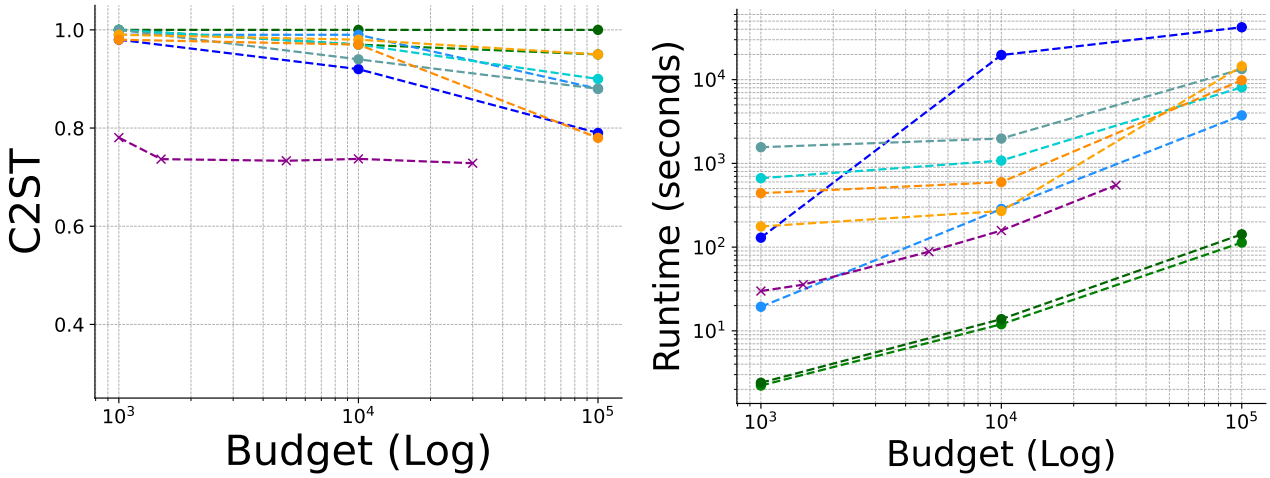


Figure 17: T.4: SLCP Distractors. From left to right: `C2ST` vs. budget, and runtime vs. budget.

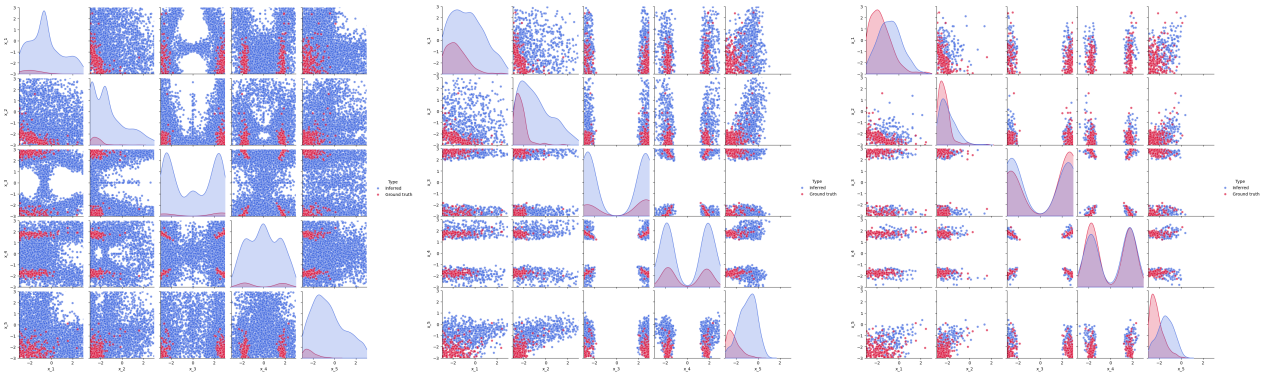


Figure 18: T.3: SLCP. From left to right: total samples, accepted samples, and selected samples by R2OMC.

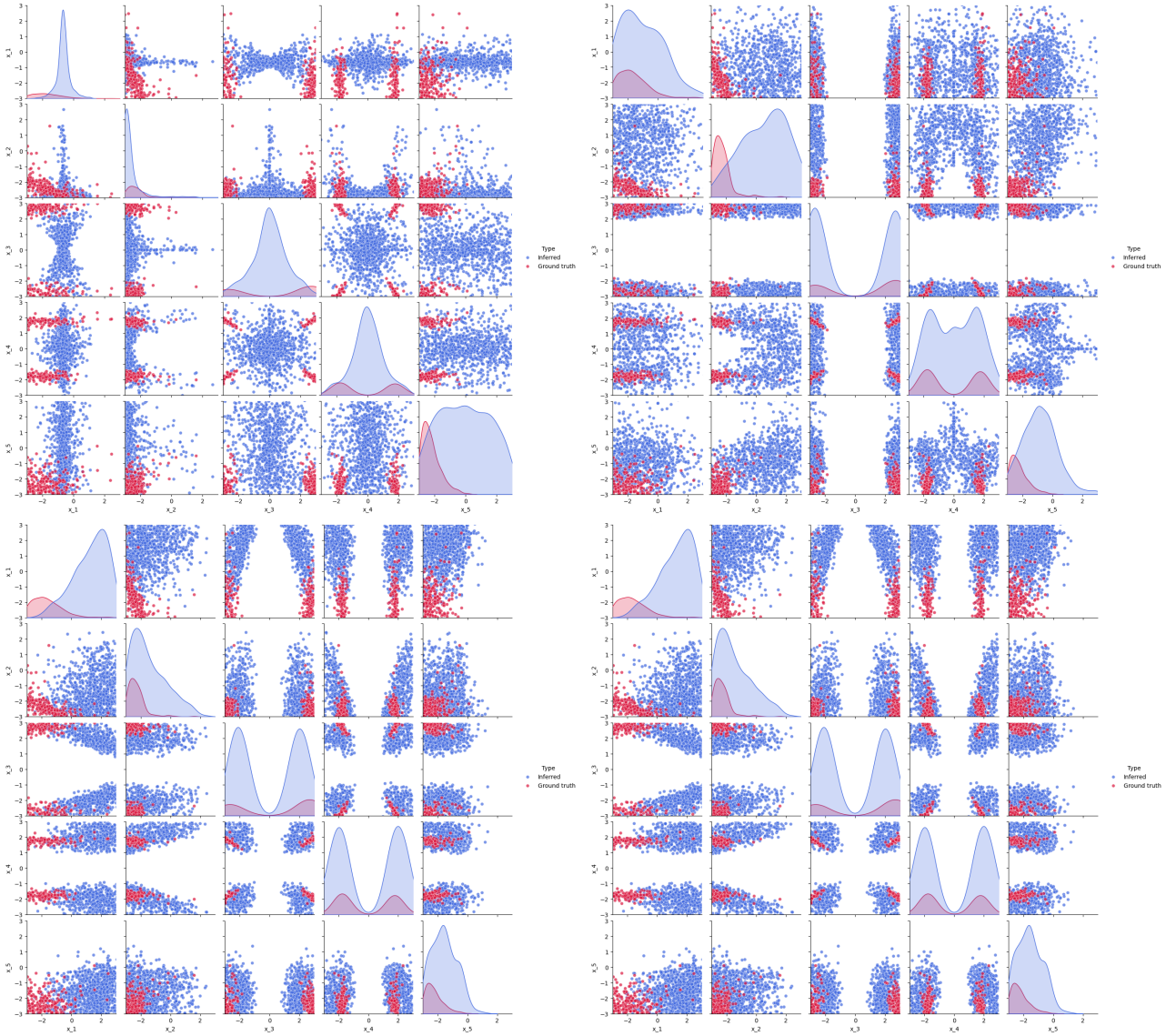
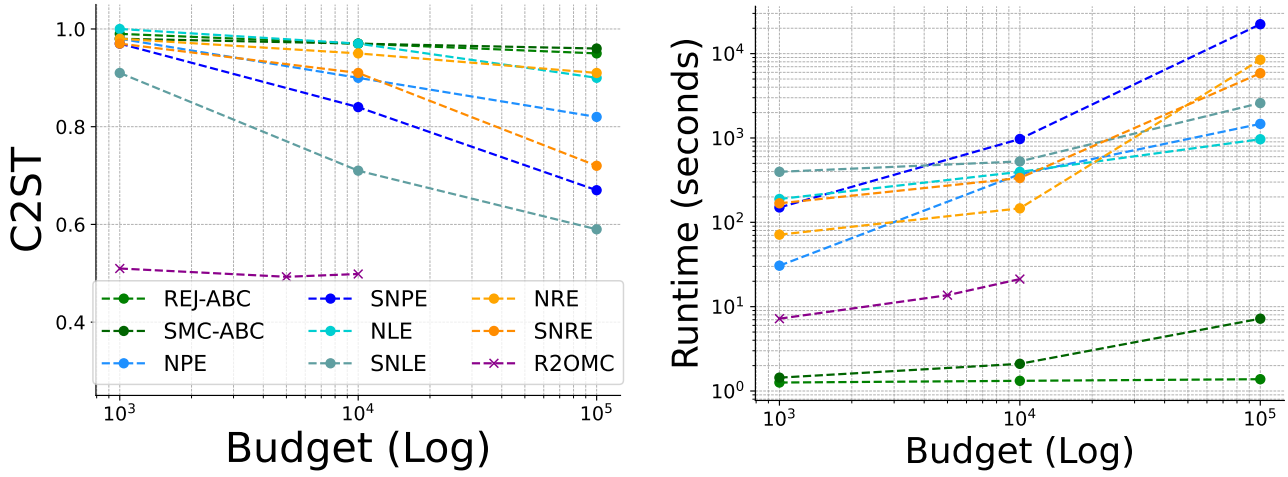


Figure 19: T.3: SLCP. From left to right: total samples, accepted samples, and selected samples by R2OMC.



(a) C2ST score

(b) Ground truth samples

Figure 20: T.8: Two Moons. From left to right: C2ST score, ground truth samples, and samples by R2OMC.

E Additional results

We here present some additional experiments that were not included in the main text.

E.1 MoG - Shifted Gaussian

This additional experiment uses a simulator that samples from a mixture of two Gaussians, whose means are shifted by a constant value.

Prior $\boldsymbol{\theta} \sim \mathcal{U}(-15, 15)^4$

Simulator $\mathbf{y} \sim \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta} + \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}, \mathbf{I})\right)$

Dimensionality $\boldsymbol{\theta} \in \mathbb{R}^4$, $\mathbf{y} \in \mathbb{R}^4$

We will use two separate cases: one with $N = 4$ similar observations and another with $N = 4$ different observations. In both cases, the posterior is:

$$
\begin{aligned}
p(\boldsymbol{\theta}|\{\mathbf{y}_n^o\}) &\propto p(\boldsymbol{\theta}) p(\{\mathbf{y}_n^o\}|\boldsymbol{\theta}) && (53)\\
&\propto p(\boldsymbol{\theta}) \prod_{n=1}^{N} p(\mathbf{y}_n^o|\boldsymbol{\theta}) && (54)\\
&\propto \mathcal{U}(-\mathbf{15}, \mathbf{15}) \prod_{n=1}^{N} \frac{1}{2}\left(\mathcal{N}(\mathbf{y}_n^o; \boldsymbol{\theta} + \mathbf{5}, \mathbf{I}) + \mathcal{N}(\mathbf{y}_n^o; \boldsymbol{\theta}, \mathbf{I})\right) && (55)\\
&\propto \mathcal{U}(-\mathbf{15}, \mathbf{15}) \prod_{n=1}^{N} \frac{1}{2}\left(\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \mathbf{I})\right) && (56)\\
&\propto\!\!\propto \prod_{n=1}^{N} \left(\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \mathbf{I})\right) && (57)
\end{aligned}
$$

In (54), the factorization is due to the N independently and identically distributed (iid) data points. In (56), we apply the property: $\mathcal{N}(\mathbf{y}; \boldsymbol{\theta} + \boldsymbol{\alpha}, \Sigma) = \mathcal{N}(\boldsymbol{\theta}; \mathbf{y} - \boldsymbol{\alpha}, \Sigma)$. In (57), we use that the data that we will use are sufficiently far from the boundaries so that we may ignore the boundary effects in the truncated distribution.

Case 1: In Case 1, we use $N = 4$ similar observations: $\mathbf{y}_n^o \approx (0, \ldots, 0)$ for $n \in \{1, \ldots, N\}$. The ground-truth posterior is approximately:

$$
p(\boldsymbol{\theta}|\mathbf{y}^o) \approx \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \frac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \frac{1}{N}\mathbf{I}) \tag{58}
$$

Proof.

$$
\begin{aligned}
p(\boldsymbol{\theta}|\mathbf{y}_n^o) &\propto\!\!\propto \prod_{n}^{N} \left(\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \mathbf{I})\right) && (59)\\
&\propto\!\!\propto \left(\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})\right)^N && (60)\\
&\propto\!\!\propto \sum_{n=0}^{N} \binom{N}{n} \left(\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})\right)^n \left(\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})\right)^{N-n} && (61)\\
&\propto\!\!\propto \left(\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})\right)^N + \left(\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})\right)^N && (62)\\
&\propto\!\!\propto \mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \frac{1}{N}\mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \frac{1}{N}\mathbf{I}) && (63)
\end{aligned}
$$

In (62), for $n \in \{1, \ldots, N-1\}$ the terms involve products between $\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})$ and $\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})$ so they have a much smaller magnitude compared to $\left(\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})\right)^N$ $(n = N)$ and $\mathcal{N}((\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}))^N$ $(n = 0)$. This allows us to approximate the expression by retaining only these two dominant terms. In (63), we use the property that raising a Gaussian distribution to the power N results in a Gaussian with the same mean and a covariance scaled by $\frac{1}{N}$.

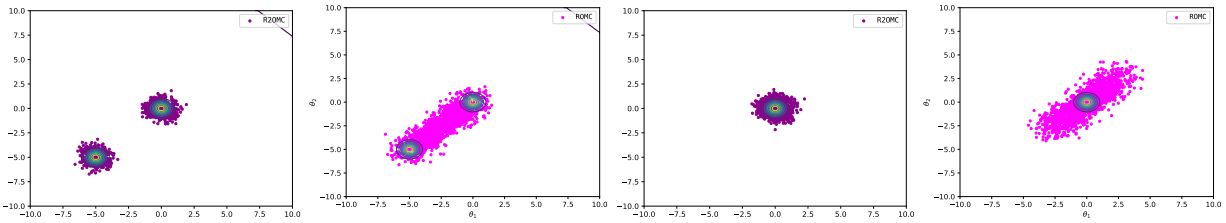


Figure 21: From left to right: R2OMC on Case 1, Naive-ROMC on Case 1, R2OMC on Case 2, and Naive-ROMC on Case 2.

Case 2: In Case 2, we use an even number of N observations that are split in two categories, half around $(0, \ldots, 0)$, i.e., $\mathbf{y}_n^o \approx (0, \ldots, 0)$ for $n \in \{1, \ldots, N/2\}$ and the other half around $(5, \ldots, 5)$, i.e., $\mathbf{y}_n^o \approx (5, \ldots, 5)$ for $n \in \{N/2+1, \ldots, N\}$. The posterior, for values of $\boldsymbol{\theta}$ sufficienty far from the boundary, is:

$$p(\boldsymbol{\theta}|\mathbf{y}^o) \approx \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \frac{1}{N}\mathbf{I}) \tag{64}$$

Proof.

$$p(\boldsymbol{\theta}|\{\mathbf{y}_k^o\}) \propto \prod_n^N (\mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o - \mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{y}_n^o, \mathbf{I})) \tag{65}$$

$$\propto \prod_{n=1}^{N/2} (\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})) \prod_{n=N/2+1}^{N} (\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I})) \tag{66}$$

$$\propto \prod_{n=1}^{N/2} (\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})) \prod_{n=1}^{N/2} (\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}) + \mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I})) \tag{67}$$

$$\propto \left((\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I}))^{N/2} + (\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}))^{N/2}\right) \left((\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I}))^{N/2} + (\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I}))^{N/2}\right) \tag{68}$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})^N \tag{69}$$

$$\propto \mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \frac{1}{N}\mathbf{I}) \tag{70}$$

In (68), we keep only the dominant terms, as in Case 1. In (69), we use that terms that involve combinations of $\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})^{N/2}$, $\mathcal{N}(\boldsymbol{\theta}; -\mathbf{5}, \mathbf{I})^{N/2}$ and $\mathcal{N}(\boldsymbol{\theta}; \mathbf{5}, \mathbf{I})^{N/2}$ have significantly smaller magnitudes compared to the dominant ones; $\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})^{N/2}\mathcal{N}(\boldsymbol{\theta}; \mathbf{0}, \mathbf{I})^{N/2}$. In (70), we use the property that raising a Gaussian distribution to the power N results in a Gaussian with the same mean and a covariance scaled by $\frac{1}{N}$.

Results. Figure 21 demonstrates that R2OMC aligns closely with the ground truth posterior, correctly capturing the two modes in Case 1 and the single mode in Case 2. In contrast, both ROMC struggles with multiple observations, resulting in their samples being distributed along the 45° line.